DAYANANDA SAGAR UNIVERSITY
School of Basic and Applied Sciences

Investigating the Impact of Rutin-Encapsulated PEG-PLGA Nanoparticles on Autophagy and Metabolic Processes in Cancer

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Declaration by Student

I, Manohari M Kulkarni (BAS21BG0023), hereby declare that, I have personally carried out the work presented in the dissertation entitled,

Investigating the Impact of Rutin-Encapsulated PEG-PLGA Nanoparticles on Autophagy and Metabolic Processes in Cancer

No part of the dissertation has been submitted for the award of any other degree or diploma prior to this date.

Date: 28th May 2024

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Certified that this dissertation work titled Investigating the Impact of Rutin-Encapsulated PEG-PLGA Nanoparticles on Autophagy and Metabolic Processes in Cancer is the bonafide work of Ms. Manohari M Kulkarni (BAS21BG0023), who carried out the work under my supervision. This is to further certify, that to the best of my knowledge the work reported herein does not form part of any other dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Cancer is a complex disease characterized by uncontrolled cell growth and proliferation, leading to the formation of malignant tumors. Despite significant advancements in treatment, it remains a substantial global health burden. This work focuses on the dysregulation of autophagy and metabolic pathways in cancer cells, highlighting their potential as therapeutic targets. Autophagy, a cellular recycling process, and metabolism, the conversion of nutrients into energy, are crucial for maintaining cellular health. However, their dysregulation in normal cells can lead to tumor formation and growth, therapy resistance, and metastasis in cancer. This thesis explores the synthesis of rutin-loaded nanoparticles using the solvent method. Rutin, a natural flavonoid known for its antimicrobial, anti-inflammatory, anticancer, and antidiabetic properties, is used as the drug to be loaded into PEG-PLGA nanoparticles. Field emission scanning electron microscopy (FESEM) was employed to analyze the surface morphology and determine particle size, with a focus on achieving uniform nanometer-scale particles. Additionally, molecular docking simulations are performed to compare the binding affinity of rutin with FDA-approved inhibitors of key proteins involved in the initiation of autophagy and metabolism. This comparative study aims to elucidate rutin's potential as an inhibitor by identifying its interactions with these critical proteins. The study also investigates the effects of rutin-loaded nanoparticles on cancer cell viability, proliferation, and apoptosis in vitro, and evaluates their potential in vivo efficacy and safety profiles. By targeting autophagy and metabolism, we aim to normalize cellular processes that are hijacked by cancer. The potential benefits of this approach include improved targeting of cancer cells with reduced side effects on healthy tissues and the utilization of a potentially less toxic natural product compared to traditional treatments. The exploration of rutin nanoparticles holds promise for the development of more effective and highly specific cancer therapies.

CHAPTER 1: INTRODUCTION

Globally, cancer contributes to one out of every six deaths, making it the second leading cause of premature death after cardiovascular diseases (Mahase, E., 2019; Bray, F., et al., 2021). Cancer is projected to overtake cardiovascular diseases and emerge as the primary cause of mortality globally in the coming decades. Presently, it holds the rank of being the foremost cause of death in 57 nations and ranks as the second most prevalent cause of mortality across 70 countries in the world (Bizuayehu, H. M., et al., 2024). Cancer is a multigenic and multicellular disease that can arise from all cell types and organs with a multi-factorial etiology. In the past decade, there has been a considerable progress towards treatment and understanding of cancer and coupled with enhancements in early detection and the diversification of the treatment options, many cancers have become curable (Baskar, R., et al., 2012). Despite all the advancements in cancer treatments, the complexities inherent in cancer biology, combined with the rising challenge of resistance to traditional treatments, emphasize the critical necessity for innovative and potent therapeutic approaches. The conventional therapeutic approaches used in cancer treatment include surgery, chemotherapy, radiation therapy, targeted therapy, immunotherapy and hormone therapy (Debela, D. T., et al., 2021). These approaches have undoubtedly made significant strides in extending survival and improving quality of life for many patients; however, these are often accompanied by many side effects and limited efficacy, especially in advanced stages of cancer (Debela, D. T., et al., 2021). Hence, there is an urgent need to explore alternative approaches that can overcome these obstacles.

Recent efforts have been made to address the limitations of existing therapeutic approaches using nanoparticles (Gavas, S., et al., 2021). In recent years, nanotechnology has emerged as a promising frontier in cancer therapy, offering the potential to revolutionize drug delivery and enhance the efficacy of anticancer agents (Yan Lee, P., & KY Wong, K., 2011; Gavas, S., et al., 202). Nanoparticle-based drug delivery systems have unique advantages, including the ability to target specific cells or tissues, improve drug solubility and stability, and minimize systemic toxicity (Mishra, D., et al., 2013). Among the diverse array of nanoparticle types, polyethylene glycol (PEG)-poly (lactic-co-glycolic acid) (PLGA) nanoparticles have emerged as an effective carrier system for delivering therapeutic agents to cancer cells (Locatelli, E., & Comes Franchini, M., 2012)

Simultaneously, natural compounds derived from plants i.e., phytochemicals, have attracted increasing interest as potential anticancer agents, owing to their diverse pharmacological properties and relatively low toxicity profiles (Mollakhalili Meybodi, N., et al., 2017; Hosseini, A., & Ghorbani, A., 2015). Rutin, a flavonoid glycoside found abundantly in various fruits and vegetables, and herbs, has emerged as a promising candidate for cancer therapy (Batra, P., & Sharma, A. K.,2013). Rutin is known to exhibit a broad spectrum of biological activities, such as antioxidant, anti-inflammatory, and anticancer effects among other activities (Imani, A., 2021; Yang, J., 2008; Yoo, H., 2014). Most importantly several studies suggest that Rutin may modulate key cellular processes implicated in cancer progression, including autophagy and metabolic pathways, which are focused on in this study (Park, M. 2016; Zhou, M., 2021).

Autophagy, a highly conserved cellular process, is essential for maintaining cellular homeostasis by facilitating the degradation and recycling of damaged organelles and macro-molecules. This process is crucial for cell survival under various stressful conditions such as energy depletion, nutrient starvation, oxidative stress and hypoxia. Dysregulation of autophagy plays a crucial role in pathogenesis of cancer, where it can have both pro-survival and pro-death effects on cancer cells depending on the cellular context (Wen, W., et al., 2024). In cancer, abnormal autophagy regulation contributes to tumor development, progression, and resistance to therapy (Das, C. K., 2018). On one hand, autophagy can promote cancer cell survival by enabling the recycling of cellular components and providing metabolic substrates to sustain tumour growth. On the other hand, excessive autophagy activation can lead to autophagic death or sensitization of cancer cell to apoptosis-inducing agents, thereby inhibiting tumour growth.

Metabolic reprogramming is another hallmark feature of cancer cells, there is alterations in energy metabolism and nutrient utilization to support their rapid proliferation and survival in harsh microenvironments. Cancer cells show increased glycolysis, even in the presence of oxygen (the Warburg effect), as well as alterations in lipid metabolism, amino acid metabolism, and mitochondrial function. These metabolic adaptations not only provide energy and building blocks for tumour growth but also offers resistance to cell death and promotes cancer cell invasiveness and metastasis (Di Gregorio, J., et al., 2022; Schiliro, C., & Firestein, B. L., 2021).

Studies have shown that Rutin can regulate autophagy by interfering in key signaling pathways involved in autophagy induction and execution, such as the mTOR (mechanistic target of Rapamycin) pathway (Dong, R., et al., 2023) and the AMPK (AMP-activated protein kinase) pathway (Enogieru, A. B., et al., 2021).

Wang, S., 2021, explores the potential of nanotechnology to enhance the effectiveness of phytochemicals in cancer therapy. Nanoparticle delivery systems offer several advantages over direct administration of phytochemicals to cancer cell lines. Firstly, phytochemicals often suffer from poor aqueous solubility and low bioavailability, which can limit their absorption and distribution within the body. Encapsulation of phytochemicals into nanoparticles addresses these challenges by improving their solubility and stability, thereby enhancing their bioavailability and systemic circulation. Additionally, nanoparticles can protect phytochemicals from enzymatic degradation and premature metabolism, allowing for sustained release and prolonged exposure of cancer cells to therapeutic agents. Furthermore, nanoparticles can be engineered to target specific cellular receptors or tumour microenvironments, thereby enhancing the selective uptake of phytochemicals by cancer cells and minimizing off-target effects on healthy tissues.

OBJECTIVES

This study is done to comprehensively explore the potential of Rutin-encapsulated PEG-PLGA nanoparticles as a novel therapeutic approach for cancer treatment. The objectives include:

a) Preparation of Rutin-loaded PEG-PLGA nanoparticles using the solvent-antisolvent method.

b) Using molecular docking to investigate the inhibitory potential of Rutin on proteins involved in the autophagy and metabolism pathways and compare it with the known, FDA approved inhibitors of the selected proteins

c) Elucidating the effects of Rutin-encapsulated nanoparticles on cancer cells by conducting cell toxicity studies on cancer cell lines

CHAPTER 2: LITERATURE REVIEW

2.1 CANCER

Cancer is a group of diseases that cause cells in the body to change and grow out of control, with each type having unique features. (Rajput Jayesh ,2023). It can arise in any part of the body, from skin to any internal organ and manifest itself in a large number of ways. Some cancers grow slowly and may not cause symptoms for years, while others can spread rapidly, spreading to other parts of the body(metastasis) (Friberg, S., & Mattson, S.,1997). Fast-growing tumors (acute leukemias, nonseminomatous testicular tumors, anaplastic thyroid carcinomas, and some pediatric tumors) will “surface” in a year or two. Slow-growing tumors (breast, prostate, colon, and several others) are said to require several years or even decades to reach detection level (Friberg, S., & Mattson, S.,1997).

2.1.1 Causes of cancer

The causes of cancer are complex and diverse, often involving a combination of genetic, environmental and lifestyle factors. Wu et al. (2018) categorized cancer risk factors on their biological nature, modifiability, and their use in literature. Firstly, there are unmodifiable intrinsic risks, which include unavoidable spontaneous mutations arising from random errors in DNA replication. These errors occur at random rates and contribute widely to cancer occurrence across populations and its prevalence over time. Secondly, non-intrinsic risks that are exogenous include modifiable, external factors such as carcinogens, viruses, radiation and lifestyle factors like smoking, nutrient imbalance and lack of physical activity. Additionally, endogenous factors such as immune response, biological aging, genetic susceptibility and hormone levels influence cell growth and genome integrity and are partially modifiable.

2.1.2 Carcinogenesis

Carcinogenesis is a process by which normal cells transform into cancer cells. It involves three stages: Initiation, Promotion and Progression, driven by genetic mutations, environmental factors and cellular interactions (Macaluso, M., et al., 2003). The uncertainty between “cause” and “explanation” complicates understanding carcinogenesis. Although external cancer agents are identified, interventions like reducing exposure or treating linked conditions are not to be considered as cures (Sonnenschein, C., & Soto, A. M., 2008).

2.1.3 Hallmarks of cancer (Weinberg, R. A., & Hanahan, D.,2000)

Weinberg, R. A., & Hanahan, D.,2000 proposed the "Hallmarks of Cancer", which provides a general framework for understanding the biology of cancer cells. This paper lays the groundwork for identifying novel therapeutic targets and developing more effective strategies for cancer diagnosis, treatment and prevention. These hallmarks include-

1) Sustained proliferative signaling: Normal cells require external signals to initiate proliferation, but cancer cells develop mechanisms to promote uncontrolled growth, often through mutations in genes that regulate cell cycle progression.

2) Evasion of growth suppressors: Healthy cells possess mechanisms that inhibit excessive proliferation, such as tumor suppressor genes that regulate cell cycle checkpoints. Whereas cancer cells, however, can bypass these regulatory mechanisms, allowing them to continue multiplying despite signals that would inhibit growth.

3) Resistance to cell death: Apoptosis, or programmed cell death, is a natural process that eliminates damaged or unwanted cells. In healthy cells, apoptosis serves as a protective mechanism to eliminate cells that are damaged, mutated, or no longer needed. This process ensures tissue homeostasis and stops the proliferation of abnormal cells.

However, cancer cells disrupt apoptosis through various mechanisms. One common way is by dysregulating key signaling pathways involved in apoptosis, such as the intrinsic and extrinsic pathways.

4) Limitless replicative potential: Non-cancer cells die after certain number of divisions, due to presence of telomeres (protective caps on the end of chromosomes that shorten with each cycle of cell division). Cancer cells bypass this barrier by acquiring genetic alterations and replicate indefinitely overriding senescence. And in order to maintain telomere length, cancer cells activate an enzyme called "telomerase", a specialized reverse transcriptase enzyme that adds repetitive DNA sequences to the ends of chromosomes, opposing telomere shortening. Telomerase is normally inactive in most somatic cells but is upregulated in a majority of cancer cells.

5) Sustained Angiogenesis: Angiogenesis is the process by which new blood vessels are formed from pre-existing vasculature. In normal conditions, this process is tightly regulated and occurs primarily during embryonic development, wound healing and tissue repair. However, in the context of cancer, this process becomes dysregulated and is exploited.

Tumor cells require a steady blood supply to fuel their growth and survival, by sustaining the process of angiogenesis, cancer cells ensure a stable supply of oxygen and nutrients.

6) Tissue invasion and metastasis: Cancer cells have the ability to invade surrounding healthy tissues and spread to distant sites in the body, a process known as "metastasis". The metastatic cascade is a complex and multi-step process involving several key stages like- Local invasion, Intravasation, Survival in circulation, Extravasation and Colonization.

- Deregulated cellular energetics is also a hallmark of cancer with cancer cells exhibiting altered metabolism compared to normal cells. The Warburg Effect describes a metabolic phenomenon wherein cancer cells preferentially use glycolysis for energy production, even in the presence of oxygen. The most important advantage of Warburg effect is rapid ATP generation, this is because glycolysis generates metabolic intermediates, such as lactate and pyruvate, that can be diverted into biosynthetic pathways to support the production of macromolecules essential for all the cellular processes. This helps cancer cells to maintain energy production even in the regions with limited or no oxygen supply.

2.2 AUTOPHAGY

Autophagy, derived from the Greek words for "self" and "eating", is an essential cellular process involved in the degradation, recycling and removal of cellular components to sustain cellular metabolism (Mathew, R., et al., 2007). It serves as a quality control mechanism, helping to maintain cellular homeostasis. In autophagy, cytoplasmic macromolecules, damaged organelles, protein aggregates, and intracellular pathogens are delivered to lysosomes, and digested by lysosomal hydrolases, which generates various molecules such as nucleotides, amino acids, sugars and ATP and finally recycled into cytosol (Li, X., et al., 2020).

2.2.1 Molecular basis of autophagy

When cells encounter stressors such as nutrient starvation, hypoxia, pathogen infection or pathological conditions such as neurodegenerative diseases and hereditary myopathies, autophagy takes place, resulting in either adaptation and survival or death (Kondo, Y., et al., 2005). Autophagy is often functionally related to apoptosis on a molecular level. It has been proposed as an important non-apoptotic cell death mechanism (Edinger, A.L., et al.,2004).

Maiuri, M. C., et al., 2007, explains that in some contexts, autophagy can promote cell survival but under certain conditions, excessive or dysregulated autophagy can lead to cell death, this is called as “autophagic cell death”. There are common regulatory mechanisms between autophagy and apoptosis pathways. For instance, certain signaling molecules such as members of the Bcl-2 protein family, can modulate both autophagy and apoptosis. Some proteins involved in autophagy, such as Beclin-1, can interact with the components of the apoptosis pathway, influencing cell fate decisions. Maiuri, M. C., et al., 2007 further suggests that modulating autophagy to enhance cancer cell death or sensitizing cancer cells to apoptosis-inducing therapies holds a promising potential to combat tumor progression.

2.2.2 Types of Autophagy

There are three defined types of autophagy: macro-autophagy, micro-autophagy and chaperone-mediated autophagy. This classification is done based on multiple routes for autophagic degradation; 1) macro-autophagy (often calls as autophagy) in which cytoplasmic contents are encompassed in double or multimembrane autophagosomes, and are then delivered to lysosome for elimination (Arstila, A. U., & Trump, B. F.,1968). 2) micro autophagy, where cytoplasm is directly engulfed by the lysosomal membrane (Ahlberg, J., et al., 1982). 3) chaperone-mediated autophagy in which proteins with a specific signal sequence are transported to the lysosomal membrane by a receptor-mediated process (Cuervo, A. M., & Dice, J. F., 1996). Of these routes, only macro autophagy has been associated with type 2 cell death (Debnath, J.,2005).

2.2.3 Autophagy-related (ATG) genes.

To understand molecular machinery of autophagy, it is really important to understand the function of ATG genes. To date, independent genetic screens have identified 30-32 ATG genes, which are involved in various subtypes of macro-autophagy, including starvation-induced autophagy, the cytoplasm-to-vacuole targeting pathway and pexophagy. Many of these genes are conserved in slime mold, plants, worms, flies and mammals. This provides a basis for studying autophagy in model organisms that represent different branches of the evolutionary tree. The insights from these studies in simpler organisms like yeasts and worms can often be used to understand complex organisms, including humans (Nakatogawa, H., et al., 2009).

Key ATG genes include ATG1/ULK1, which initiates autophagy (Mizushima, N., 2010); ATG5, part of the Atg12-Atg5-Atg16 complex essential for autophagosomal membrane elongation (Tanida, I., 2011); ATG7, facilitating the conjugation of Atg12 to Atg5 and lipidation of LC3/Atg8 (Fujita, N., et al., 2008); ATG8/LC3, which conjugates with phosphatidylethanolamine (PE) to form LC3-II, crucial for autophagosome formation (Huang, X., et al., 2022); ATG9, involved in membrane trafficking to autophagosome assembly sites (He, C., & Klionsky, D. J., 2007); BECN1/Beclin-1, forming a complex with PI3K to initiate autophagy (Menon, M. B., & Dhamija, S., 2018); ATG16L1, part of the Atg12-Atg5-Atg16 complex and essential for LC3 lipidation (Fujita, N., et al., 2008); and ATG4, regulating LC3 processing (Zhou, Y., et al., 2021). Additionally, while not traditionally considered an autophagy-related gene, MTORC1 plays a pivotal role in autophagy regulation, integrating signals from nutrient availability to modulate autophagy initiation through phosphorylation of key targets such as ULK1 and Beclin-1(Wang, Y., & Zhang, H.,2019).

The coordinated action of these genes supports the complex molecular machinery underlying autophagy, ensuring cellular adaptation and survival in response to diverse environmental cues and stresses.

2.2.4 The process of Autophagy

The core machinery of autophagy includes several steps that start from initiation and end at degradation. Each step is explained below-

• Initiation: Various signals, such as nutrient deprivation, oxidative stress, or the presence of damaged organelles, triggers the initiation of autophagy. These signals indicate that the cell needs to recycle its components to maintain homeostasis and survive under stress conditions (He, L., et al., 2018). The key regulator of autophagy initiation is the “mTOR pathway”. Under normal conditions, mTOR is active and inhibits autophagy. Whereas under stress, mTOR activity is inhibited. This inhibition initiates autophagy. Upon the inhibition of mTOR signaling, the ULK1 complex, composed of ULK1, FIP200, ATG13, and ATG101 is activated. The key kinase in this process is Unc-51-like autophagy activating kinase 1 (ULK1). ULK1 phosphorylates other components of the complex to activate it which leads to the nucleation of the phagopore (Wang, Y., & Zhang, H.,2019).

• Nucleation: Phagopore is a double membrane structure that surrounds the components that need to be degraded. This process is controlled by the class III phosphatidylinositol 3-kinase (PI3K) kinase (Zhang, R., et al., 2022). PI3K kinase consists of: i) Vps34 – The catalytic subunit of PI3K complex and phosphorylates phosphatidylinositol to generate phosphatidylinositol 3-phosphate (PI3P). This is a crucial signaling lipid involved in membrane trafficking and autophagosome formation. ii) Vps15 – The regulatory subunit that stabilizes the kinase activity of Vps32. It helps in targeting PI3K complex to specific cellular membranes. iii) Beclin 1- It is also a key regulatory component that interacts with Vps34 and facilitates its recruitment to phagophore assembly site. Beclin 1 also coordinates autophagy with other cellular processes like apoptosis and endocytosis. iv) ATG14 – It confers specificity to the PI3K complex and localizes it to the phagopore assembly site (Bilanges, B., et al.,2019)

• Elongation: During the elongation phase of autophagy, the phagophore expands and engulfs cytoplasmic components, preparing to form a fully enclosed autophagosome. This process involves two main ubiquitin-like conjugation systems: a) The Atg12-Atg5-Atg16L1 complex- It consists of ATG12, ATG5 and ATG16L1. ATG12 is joined to ATG5 via a ubiquitin-like conjugation reaction facilitated by ATG7 and ATG10.This conjugate then includes ATG16L1 to form a multimeric complex. This complex localizes to the outer membrane of the expanding phagophore and promotes its elongation. b) The microtubule-associated protein 1A/1B-light chain 3 (LC3) lipidation machinery- Initially, LC3 is synthesized as a precursor protein (proLC3), which is cleaved by the cysteine protease Atg4 to expose a C-terminal glycine residue. The cleaved LC3, known as LC3-I, is then conjugated to phosphatidylethanolamine (PE) through a ubiquitin-like conjugation reaction similar to ATG12 conjugation. This reaction is mediated by ATG7 and ATG3. The lipidated form, LC3-II, is integrated into the expanding autophagosomal membrane and it plays a crucial role in component recognition and autophagosome closure (Yang, Y., et al., 2021; Li, X., et al., 2020; Tanida, I., et al., 2004)

• Cargo recognition and Sequestration: Involves the specific targeting of particular cellular components/cargo for degradation within autophagosomes. The cell recognizes targets through various mechanisms, such as ubiquitination, specific post-translational modifications, or recognition patterns/motifs. Cargo receptors are proteins that contain LIR motifs, which enable them to bind to LC3 on the autophagosomal membrane. These cargo receptors recognize the component and delivers it to the autophagosome for degradation.

Few examples of receptors include- p62/SQSTM1, NBR1, optineurin, NDP52, and TAX1BP1. After the receptor is bound to the target, it reacts with LC3 through its LIR motif. LC3 is lipidated and localized on the autophagosomal membrane, where it acts as a marker of autophagosomes. The very specific interaction between the receptor and LC3 facilitates selective sequestration of the target component into autophagosome (Johansen, T., & Lamark, T., 2020; Adriaenssens, E., et al., 2022)

• Autophagosome maturation and Fusion with Lysosomes: After sequestration, the autophagosome undergoes maturation and acquires proteins and lipids necessary for fusion with lysosomes. This fusion involves specific proteins like Rab GTPases, SNARE proteins. Rab GTPases are small GTP-binding proteins. Several GTPases like Rab7, Rab5 are particularly involved in late-stage autophagosome maturation and fusion with lysosome. Soluble N-ethylmaleimide-sensitive factor attachment protein receptors (SNAREs) are membrane proteins that facilitate the fusion of autophagosomes with lysosomes. Once the autophagosome and lysosome come into close proximity, Rab GTPases helps to bind the membranes by interacting with specific effector proteins on both membranes. This binding process precedes membrane fusion and is crucial for bringing the autophagosome and lysosome membranes together. The SNARE proteins on the autophagosome and lysosome form a trans-SNARE complex, which drives membrane fusion between the autophagosome and lysosome. This complex undergoes a conformational change, leading to the fusion of the lipid bilayers of the autophagosome and lysosome, forming an autolysosome (Zhao, Y. G., & Zhang, H.,2019; Tian, X., et al., 2021; Homma, Y., et al., 2021)

• Degradation: Inside the autolysosome, the lysosomal enzymes break down the cellular components, which can include proteins, lipids, carbohydrates, and organelles, into smaller molecules. These molecules are then released back into the cytoplasm, where they can be reused by the cell to synthesize new components or to generate energy. The presence of both the autophagosomal membrane and lysosomal enzymes in the autolysosome ensures the efficient degradation of cargo while maintaining the integrity of the cellular components. This process is vital for cellular homeostasis, as it allows cells to remove damaged organelles, misfolded proteins, and other unnecessary or harmful components, thus contributing to overall cell health and function (Mahapatra, K. K., et al., 2021; Buratta, S., et al., 2020)

2.2.5 Autophagy Regulation

Autophagy is tightly regulated by a complex network of signaling pathways that integrate various cellular stressors and metabolic cues. Key regulators include nutrient-sensing pathways, such as the mTOR complex 1 (mTORC1) and AMP-activated protein kinase (AMPK), as well as stress-responsive kinases and transcription factors.

- mTOR Signaling: mTORC1 serves as a master regulator of autophagy, inhibiting its initiation in response to growth factors and nutrient availability. Upon nutrient deprivation, mTORC1 is inhibited, leading to the activation of autophagy through the ULK1 complex (Rabanal-Ruiz, Y., & Korolchuk, V. I., 2018)

- AMPK Pathway: A) AMPK serves as a cellular energy sensor, detecting changes in the AMP:ATP ratio. When cellular ATP levels decrease (indicating low energy availability) or AMP levels increase (indicating energy depletion), AMPK is activated. B) AMPK directly phosphorylates ULK1, a key component of the autophagy initiation complex, at multiple sites, including Ser317, Ser555, and Ser777. C) AMPK inhibits the activity of mTORC1, a major negative regulator of autophagy. mTORC1 is sensitive to cellular energy status and is inhibited by AMPK-mediated phosphorylation of regulatory proteins such as TSC2 (tuberous sclerosis complex 2) and Raptor (regulatory-associated protein of mTOR) (Lin, S. C., & Hardie, D. G., 2018; Kim, J., et al., 2011; Gwinn, D. M., & Shaw, R. J.,2010)

- Transcriptional Regulation: Autophagy is also modulated at the transcriptional level by various transcription factors, including TFEB (transcription factor EB) and other members of the MiT/TFE (Microphthalmia/Transcription Factor E) family. TFEB is a master regulator of lysosomal and autophagy gene expression. When cells experience nutrient deprivation or other stresses, TFEB translocates to the nucleus and activates the expression of genes involved in lysosome biogenesis, autophagy initiation, and autophagosome-lysosome fusion. This results in the upregulation of the autophagy-lysosome pathway, promoting cellular adaptation to stress and the clearance of damaged organelles and protein aggregates (Slade, L., & Pulinilkunnil, T., 2017)

p53, a tumor suppressor protein, can induce autophagy in response to various stress signals, including DNA damage and oxidative stress. FOXO transcription factors are involved in the regulation of diverse cellular processes, including autophagy, in response to nutrient deprivation and oxidative stress. NF-κB, a key regulator of inflammation and immunity, can

both positively and negatively regulate autophagy depending on the cellular context and signaling pathways involved (Yang, Y., et al., 2020; Webb, A. E., & Brunet, A.,2014; Trocoli, A., & Djavaheri-Mergny, M.,2011)

2.2.6 Proteins involved in Autophagy Induction

- mTORC1 (Mammalian Target of Rapamycin Complex 1):

mTORC1 is a key regulator of cellular metabolism, growth, and survival, integrating various environmental cues, including nutrient availability, energy status, and growth factors. In nutrient-rich conditions, active mTORC1 suppresses autophagy, inhibiting the initiation of autophagosome formation. This inhibition occurs through phosphorylation of key autophagy regulators, including ULK1 and ATG13, which are critical for autophagy induction. mTORC1 exerts its inhibitory effect on autophagy by phosphorylating ULK1 at multiple sites, including Ser757, leading to the sequestration of ULK1 and ATG13 in an inactive complex and preventing the initiation of autophagy.

Conversely, under nutrient-deprived or stress conditions, mTORC1 activity is inhibited, relieving its suppression of autophagy. This inhibition can occur through various mechanisms, including decreased amino acid levels, energy depletion, or activation of stress-responsive signaling pathways. Inhibition of mTORC1 results in the dephosphorylation and activation of ULK1, triggering the initiation of autophagy. Activated ULK1 phosphorylates downstream autophagy effectors, such as Beclin-1 and ATG14, promoting the nucleation of autophagosomes and the recruitment of autophagy-related proteins to the phagophore assembly site (Dossou, A. S., & Basu, A.,2019)

- ULK1 (Unc-51 Like Autophagy Activating Kinase 1):

ULK1 is a serine/threonine kinase that plays a central role in the initiation of autophagy, acting as a key regulator of autophagy induction and autophagosome formation. In nutrient-rich conditions, ULK1 activity is suppressed by mTORC1-mediated phosphorylation at multiple sites, including Ser757, Ser637, and Ser758. Phosphorylation at these sites inhibits ULK1 kinase activity and prevents its association with other autophagy initiation factors, such as ATG13 and FIP200, thereby blocking autophagy induction.

In contrast, during nutrient deprivation or stress conditions, inhibition of mTORC1 leads to the dephosphorylation and activation of ULK1, enabling its association with ATG13 and FIP200 to form the ULK1 complex. Activated ULK1 phosphorylates downstream targets,

including Beclin-1 and ATG14, promoting the nucleation of the autophagosomal membrane and the initiation of autophagy. ULK1 also regulates other aspects of autophagy, such as autophagosome maturation and fusion with lysosomes, through interactions with additional autophagy effectors and regulatory proteins.

In summary, mTORC1 and ULK1 represent key regulators of autophagy induction, orchestrating the cellular response to nutrient availability, energy status, and stress conditions. The dynamic interplay between mTORC1-mediated inhibition and ULK1-mediated activation finely tunes the initiation of autophagy, balancing cellular homeostasis, and adaptation to environmental challenges (Wirth, M., et al., 2013)

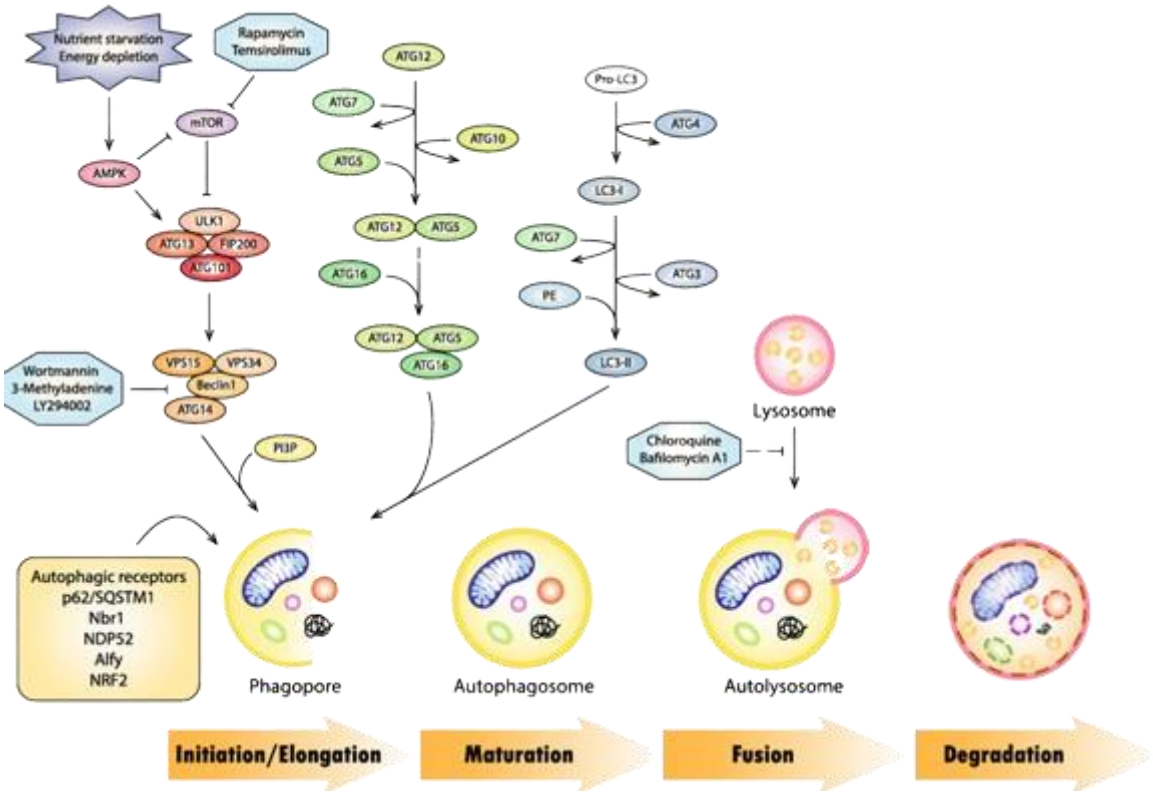


Fig 2.1 Mechanism of autophagy (Image source: Lim, S. M., et al., 2021)

2.3 METABOLISM

Altered metabolism is one of the hallmarks of cancer cells, which supports their rapid growth and survival. In healthy cells, energy production predominantly occurs through oxidative phosphorylation within the mitochondria, a process known for its efficiency in generating adenosine triphosphate (ATP) through the sequential oxidation of substrates in the presence of oxygen. This process ensures an ample supply of cellular energy required for various physiological functions, maintaining cellular homeostasis and viability (Senior, A. E., 1988).

On the contrary, cancer cells often undergo a metabolic alteration, known as the Warburg effect or aerobic glycolysis, wherein they prioritize glycolysis as the primary mechanism for ATP synthesis, even in the presence of oxygen (Hirschey, M. D., et al., 2015).

Under physiological conditions, glucose serves as the main substrate for producing energy, undergoing glycolysis to yield pyruvate, which enters the TCA cycle within the mitochondria. During glycolysis, a single molecule of glucose is enzymatically cleaved into two molecules of pyruvate, generating a net of two molecules of ATP and reducing equivalents in the form of nicotinamide adenine dinucleotide (NADH). Pyruvate then undergoes oxidative decarboxylation to form acetyl-CoA, which enters the TCA cycle to fuel the electron transport chain (ETC) for ATP synthesis through oxidative phosphorylation. This process occurs in aerobic conditions, where oxygen serves as the terminal electron acceptor, facilitating the efficient generation of ATP (Mayes, P. A., & Bender, D. A., 2003).

The cancer cells exhibit glycolytic metabolism, irrespective of oxygen availability (Warburg effect). This metabolic shift is characterized by increased glucose uptake and utilization, with a substantial portion of glucose-derived carbon flux diverted towards lactate production, even in the presence of oxygen- a phenomenon called aerobic glycolysis. Despite being less efficient in terms of ATP yield per glucose molecule compared to oxidative phosphorylation, aerobic glycolysis provides several advantages to cancer cells, including rapid ATP generation, production of metabolic intermediates for biomass synthesis, and adaptation to hypoxic microenvironments within solid tumors. Furthermore, glycolytic intermediates can be used in other pathways, such as the pentose phosphate pathway (PPP) and serine/glycine biosynthesis, to support nucleotide and lipid synthesis, thereby sustaining the proliferative capacity of cancer cells (Bartrons, R., & Caro, J., 2007)

The dysregulated metabolism observed in cancer cells arises from various genetic and epigenetic alterations that collectively alter cellular signaling networks and metabolic pathways to help tumorigenesis and disease progression. Key regulators of metabolic reprogramming in cancer include oncogenes such as c-Myc, which transcriptionally upregulates glycolytic enzymes and glucose transporters to promote aerobic glycolysis, and tumor suppressors like p53, which modulate metabolic pathways in response to cellular stressors. Additionally, dysregulation of signaling pathways such as phosphoinositide 3-kinase (PI3K)/Akt/mammalian target of rapamycin (mTOR) and AMP-activated protein

kinase (AMPK) pathway further increases the metabolic rewiring in cancer cells, sustaining their proliferative and survival advantages (Marbaniang, C., & Kma, L.,2018).

2.3.1 The Warburg Effect

The Warburg effect was first described by Otto Warburg in the 1920s, who observed that cancer cells exhibit high rates of glycolysis followed by lactic acid fermentation, even under aerobic conditions. This metabolic phenotype is characterized by increased glucose uptake and lactate production, despite the availability of oxygen. While less efficient in terms of ATP production per glucose molecule, glycolysis provides cancer cells with several advantages such as- rapid ATP generation, acidification of the tumor microenvironment which promotes tumor invasion and metastasis while suppressing immune responses and diversion of intermediates into biosynthetic pathways to produce macromolecules essential for cell growth (Vaupel, P., et al., 2019).

2.3.2 Mechanisms of Dysregulated Metabolism in cancer cells

Several mechanisms contribute to the Warburg effect and overall metabolic reprogramming in cancer:

a) Oncogene Activation: Oncogenes such as MYC, Ras, and components of the PI3K/Akt signaling pathway have profound effects on cellular metabolism, fueling the energetic demands of proliferating cancer cells. For instance, MYC, a potent oncogenic transcription factor, directs the transcriptional activation of glycolytic enzymes and glucose transporters, promoting increased glucose uptake and metabolism. Similarly, Ras and PI3K/Akt signaling pathways converge on key metabolic nodes to stimulate glycolysis, enhance nutrient uptake, and promote anabolic processes, collectively driving the metabolic switch observed in cancer cells (Stine, Z. E., et al., 2015; Lien, E. C., et al., 2016).

b) Tumor Suppressor Inactivation: In cancer, the inactivation of tumor suppressor genes, such as p53, disrupts the finely tuned regulation of cellular metabolism. P53, a critical guardian of genomic integrity, also plays a pivotal role in modulating metabolic homeostasis. However, frequent mutations or inhibition of p53 in cancer cells result in dysregulated metabolism, characterized by increased glycolysis and aberrant mitochondrial function. Loss of p53 function not only enhances glycolytic flux but also disrupts mitochondrial respiration, leading to a

shift towards aerobic glycolysis and decreased oxidative phosphorylation. This metabolic rewiring promotes the metabolic advantages by cancer cells, supporting their proliferative and survival capacities amidst challenging microenvironments (Yeung, S. J., et al., 2008; Chen, J. Q., & Russo, J., 2012).

c) Hypoxia-Inducible Factors (HIFs): In the hypoxic microenvironments characteristic of solid tumors, the activation of Hypoxia-Inducible Factors (HIFs) emerges as a critical adaptive response. Particularly, HIF-1α, a pivotal transcription factor, becomes stabilized under hypoxic conditions, giving rise to a series of molecular events that drive cancer progression. Notably, HIF-1α upregulates genes involved in diverse cellular processes, including glycolysis, angiogenesis, and metastasis. This transcriptional activation supports the Warburg effect, promoting the preferential utilization of glycolysis for ATP generation even in the presence of oxygen. Additionally, HIF-1α activation enhances the metastatic potential of cancer cells by regulating the expression of genes involved in epithelial-to-mesenchymal transition (EMT) and extracellular matrix remodeling, facilitating tumor invasion and dissemination to distant sites (Infantino, V., et al., 2021).

2.3.3 Proteins involved in cancer cell metabolism (Khan, M. A., et al., 2020)

- PFK-1 (Phosphofructokinase-1):

PFK-1 is a key regulatory enzyme in glycolysis, the metabolic pathway that converts glucose into pyruvate, generating ATP and NADH in the process. In cancer cells, PFK-1 is often upregulated or activated by various mechanisms, including genetic alterations, hypoxia-inducible factors, and oncogene signaling pathways. This increased PFK-1 activity drives glucose flux through glycolysis, even under aerobic conditions, diverting glucose-derived carbon into biosynthetic pathways to support cell growth and proliferation. Moreover, PFK-1 regulates the flow of metabolites into the pentose phosphate pathway (PPP), providing cancer cells with reducing equivalents and nucleotide precursors necessary for biomass synthesis and redox homeostasis. The dysregulation of PFK-1 in cancer contributes to the Warburg effect, enabling cancer cells to thrive in environments with limited oxygen availability while reprogramming cellular metabolism to sustain their energetic and biosynthetic needs.

- HK II (Hexokinase II):

Hexokinase II is an isoform of the enzyme hexokinase, which catalyzes the phosphorylation of glucose to glucose-6-phosphate, the first step in glycolysis. HK II plays a critical role in cancer metabolism by facilitating the rapid uptake and utilization of glucose in cancer cells. Its overexpression is frequently observed in various cancers and is associated with tumor growth, metastasis, and resistance to cell death. HK II localization to the outer mitochondrial membrane not only enhances glucose metabolism but also suppresses apoptosis by inhibiting pro-apoptotic proteins. Moreover, HK II expression is regulated by oncogenic signaling pathways, such as PI3K/AKT and MAPK/ERK, further highlighting its importance in promoting cancer cell survival and proliferation. Targeting HK II has emerged as a potential therapeutic strategy for disrupting cancer metabolism and sensitizing cancer cells to apoptosis-inducing agents.

- FASN (Fatty Acid Synthase):

FASN is a multifunctional enzyme responsible for de novo fatty acid synthesis from acetyl-CoA and malonyl-CoA. In cancer cells, FASN expression is frequently upregulated to meet the heightened demand for fatty acids, which serve as building blocks for membrane biogenesis, signaling molecules, and energy storage. Elevated FASN levels are associated with aggressive tumor phenotypes, poor prognosis, and resistance to chemotherapy. FASN upregulation in cancer is driven by various factors, including oncogenic signaling pathways (e.g., PI3K/AKT, MYC), transcriptional regulators (e.g., SREBP-1c), and hypoxia-inducible factors. In addition to its role in lipid synthesis, FASN promotes cancer cell survival and proliferation by modulating signaling pathways involved in cell growth, apoptosis resistance, and metastasis. Targeting FASN holds promise as a therapeutic approach for inhibiting cancer cell growth and sensitizing tumors to chemotherapy and targeted therapies.

- GLS (Glutaminase):

GLS is an enzyme that catalyzes the conversion of glutamine to glutamate, a key step in glutaminolysis, a metabolic pathway that fuels cancer cell growth and survival. Glutamine is a versatile nutrient that serves as a nitrogen donor for biosynthetic processes, a precursor for amino acid and nucleotide synthesis, and a substrate for energy production. GLS activity is often increased in cancer cells to support their heightened metabolic demands, including ATP generation, redox balance maintenance, and macromolecule biosynthesis. GLS upregulation is driven by oncogenic signaling pathways, such as MYC and mTOR, as well as metabolic adaptations to nutrient availability and microenvironmental stressors. Inhibition

of GLS has shown promise as a therapeutic strategy for targeting cancer metabolism and sensitizing cancer cells to nutrient deprivation-induced apoptosis.

2.4 RUTIN

Rutin, a flavonoid glycoside commonly found in various plants such as buckwheat, citrus fruits, and berries, has garnered significant attention in recent years due to its pharmacological potential. Known for its diverse range of biological activities, rutin possesses properties that make it a promising candidate in the realm of medicine and therapeutics.

Firstly, rutin exhibits potent antioxidant properties, attributed to its ability to scavenge free radicals and mitigate oxidative stress within the body (Prasad, R., & Prasad, S. B., 2019). This antioxidant activity is crucial in combating various diseases and conditions associated with oxidative damage, including cancer, cardiovascular diseases, and neurodegenerative disorders (Enogieru, A. B., et al., 2018).

Moreover, rutin has emerged as a promising anti-inflammatory agent, capable of modulating inflammatory pathways and reducing inflammation in various experimental models (Enogieru, A. B., et al., 2018). By inhibiting the production of pro-inflammatory mediators and cytokines, rutin demonstrates potential in alleviating inflammatory conditions such as arthritis, asthma, and inflammatory bowel diseases (Al-Khayri, J. M., et al., 2022).

Furthermore, numerous studies have highlighted the anticancer potential of rutin, elucidating its ability to target multiple signaling pathways involved in cancer progression and metastasis. Research conducted by Nouri et al., 2020, emphasizes the therapeutic approach of rutin in cancer treatment by targeting various signaling cascades implicated in tumor growth, angiogenesis, and metastasis. Similarly, Imani et al., 2021 delves into the molecular mechanisms underlying rutin's anticancer effects, shedding light on its ability to induce apoptosis, inhibit cell proliferation, and suppress oncogenic signaling pathways.

Previous studies have underscored rutin's efficacy in inhibiting the proliferation of cancer cells, inducing cell cycle arrest, and promoting apoptosis through various mechanisms. Additionally, rutin has been shown to inhibit angiogenesis, impair tumor invasion and metastasis, and enhance the efficacy of conventional chemotherapy agents (Farha, A. K., et al., 2022).

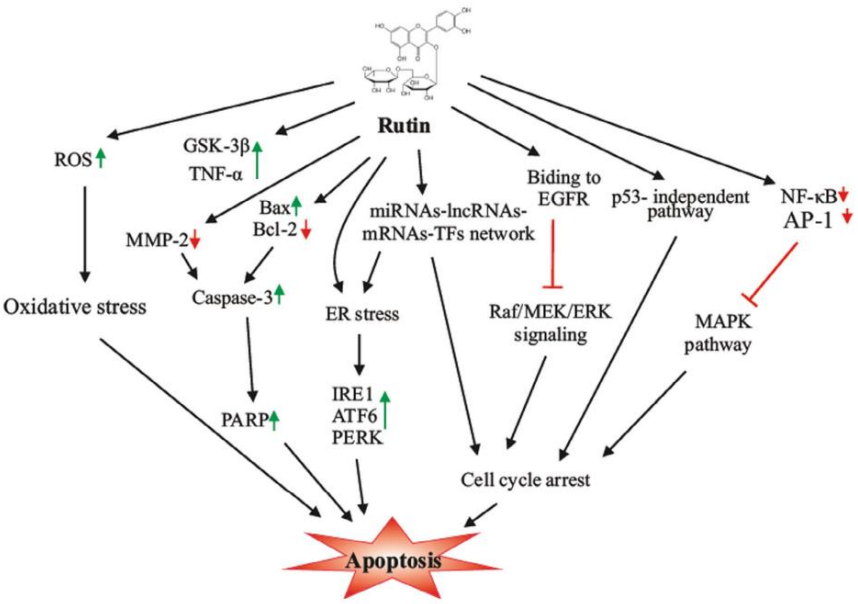


Fig 2.2: The route of action and molecular mechanism of anticancer effect of the rutin (Image source: Imani, A., et al., 2021).

2.5 NANOPARTICLES

Nanotechnology, at its core, is the science and engineering of manipulating matter at the nanoscale, typically at dimensions of 100 nanometers or smaller. Nanoparticles are frequently defined as solid, colloidal particles in the range 10–1000 nm. The drug is dissolved, entrapped, encapsulated or attached to a nanoparticle matrix. They offer a means to enhance the solubility, stability, bioavailability, and target specificity of various nutrients, phytochemicals, and natural compounds.

Mohanraj, V. J., & Chen, Y. J. T. J. O. P. R. (2006), talk about a multitude of advantages offered by nanoparticles as a drug delivery system, making them highly promising in pharmaceutical applications and cancer therapy:

a) Firstly, the particle size and surface characteristics of nanoparticles can be easily tailored to achieve both passive and active drug targeting following a close administration. This capability enables nanoparticles to accumulate selectively at the desired site of action, enhancing drug efficacy while minimizing systemic side effects to healthy tissues.

b) Secondly, nanoparticles enable controlled and sustained release of drugs during transportation and at the site of localization. By modulating drug release kinetics and altering organ distribution, nanoparticles contribute to increased therapeutic efficacy and reduced side effects, resulting in optimized pharmacokinetic profiles.

c) Moreover, the controlled release and particle degradation characteristics of nanoparticles can be readily adjusted by selecting appropriate matrix constituents. This flexibility allows for precise modulation of drug release profiles, ensuring optimal therapeutic outcomes.

d) Additionally, nanoparticles exhibit high drug loading capacity, enabling efficient incorporation of drugs into the delivery system without undergoing chemical reactions. This aspect is crucial for preserving the activity of encapsulated drugs and maximizing their therapeutic potential.

Nanoparticles come in various types, each with its unique properties and advantages. Inorganic nanoparticles, such as metal nanoparticles (e.g., gold, silver) and quantum dots, offer excellent optical and electronic properties, making them suitable for imaging and sensing applications. They also possess high stability and durability. Lipid-based nanoparticles, such as liposomes and solid lipid nanoparticles, are biocompatible and often used for drug delivery due to their ability to encapsulate both hydrophilic and hydrophobic drugs (Khan, S., & Hossain, M. K.,2022; Shirodkar, R. K., et al., 2019).

Polymeric nanoparticles, however, stand out due to their versatility and advantages over other types of nanoparticles. These nanoparticles are typically made from biodegradable and biocompatible polymers such as poly (lactic-co-glycolic acid) (PLGA), polyethylene glycol (PEG), and chitosan. One significant advantage of polymer nanoparticles is their tunable properties, including size, surface charge, and degradation rate, which can be tailored to suit specific drug delivery needs (Aghebati-Maleki, A., et al., 2020).

2.5.1 Polymeric Nanoparticles

Polymeric nanoparticles refer to nanoscale particles composed primarily of synthetic or natural polymers. They can be further categorized into different types based on their structure and composition, such as nanospheres and nanocapsules. Nanospheres are solid matrix particles where the drug molecules can be either adsorbed onto the surface or encapsulated

within the particle. On the other hand, nanocapsules are vesicular systems with a liquid core (either oil or water) surrounded by a solid material shell, acting as a reservoir for entrapped substances (Rao, J. P., & Geckeler, K. E.,2011). Polymeric nanoparticles offer high drug loading capacity and controlled release kinetics, allowing for sustained and targeted drug delivery. Moreover, they can be easily functionalized with targeting ligands or imaging agents to enhance specificity and efficacy. They also exhibit good stability in biological environments and can protect encapsulated drugs from degradation and are compatible with various routes of administration, including oral, parenteral, and topical delivery (Nahar, M., et al., 2006). They can navigate biological barriers and accumulate selectively at target sites, minimizing systemic side effects. They can be easily handled and have a cost-effective manufacturing process, making them attractive for large-scale production.

CHAPTER 3: EXPERIMENT

3.1 Materials

The drug and various other excipients and reagents used in this work are enlisted in the following Table 3.1.

Rutin	Jinay Pharmaceuticals, Pvt.Ltd, India (ChemScene LLC)
Poly (ethylene glycol) (PEG)- Poly (lactic acid-co-glycolic acid) (PLGA) [MW- 55kDa]	Nomisma HealthCare Pvt.Ltd, Vadodara, India.
Polyvinyl Alcohol (PVA) [MW: 30-70kDa]	Sigma Aldrich Pvt.Ltd, Bangalore, India
HPLC grade water (CAS no: 7732-18-5)	Qualigens, ThermoFisher Scientific India Pvt.Ltd, Mumbai, India
Dimethyl Sulfoxide	Nice Chemicals Pvt.Ltd, Kerala, India
Dichloromethane	S. D Fine Chemicals, Pvt.Ltd, Mumbai, India

3.2 Profile of the drug

3.2.1 Rutin

- CAS number: 153-18-4
- Chemical name: 2-(3,4-dihydroxyphenyl)-5,7-dihydroxy-3-[(2S,3R,4S,5S,6R)-3,4,5-trihydroxy-6-[[(2R,3R,4R,5R,6S)-3,4,5-trihydroxy-6-methyloxan-2-yl] oxymethyl] oxan-2-yl] oxychromen-4-one
- Chemical formula: $C_{27}H_{30}O_{16}$

- Structural formula:

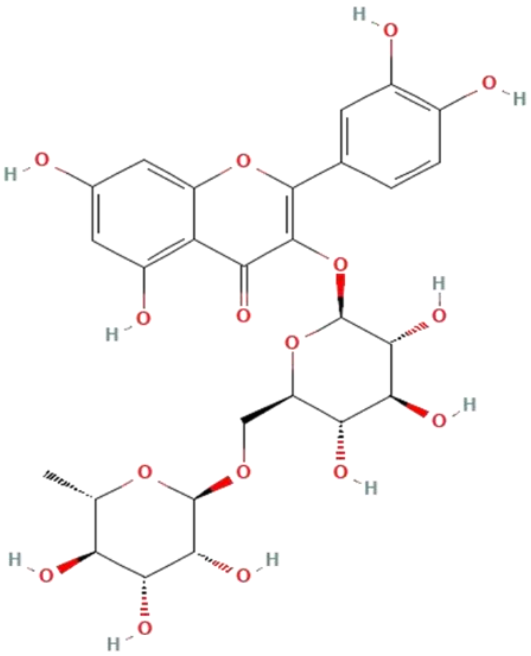


Fig 3.1 2D structure of Rutin

- **ADMET analysis of Rutin**

ADMET analysis of rutin was done to unveil its medicinal or drug-likeness perspectives. The molecular structures of rutin represented below in Fig 3.2

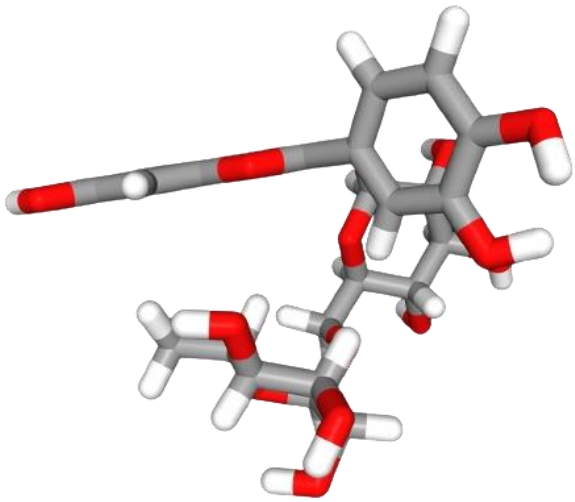


Fig 3.2 3D structure of Rutin

Rutin was first subjected to SwissADME, to infer fundamental drugs properties. The following Fig 3.3 highlights some of the crucial features of rutin.

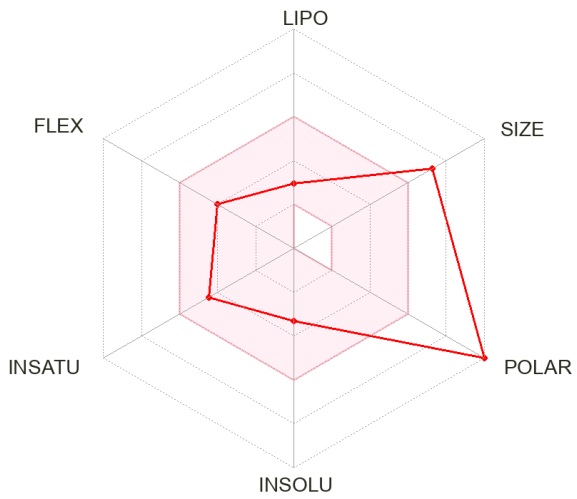


Fig 3.3 SWISSADME analysis of Rutin highlighting its features

ADMET (Absorption, Distribution, Metabolism, Excretion, and Toxicity) analyses are essential in drug discovery to evaluate a compound's pharmacokinetic and safety profiles, ensuring its suitability for therapeutic use. SWISS ADME is an online tool that facilitates the assessment of these properties by predicting various pharmacokinetic parameters, drug-likeness, and medicinal chemistry friendliness of small molecules. It provides crucial insights into a compound's absorption potential, metabolic stability, and possible toxicological risks, aiding in the selection and optimization of promising drug candidates. The collective results of SwissADME and ADMETlab 2.0 highlighted multiple pharmacokinetic aspects (Sarwar, M. F., et al., 2024). Initially, the physiochemical properties were evaluated which focused on important parameters including the molecular weight, number of heavy atoms, fraction Csp3, number of acceptor and donor hydrogen bonds and topological polar surface area (TPSA). These results are summarized in the following Table 3.2.

Table 3.2. Physiochemical properties of Cinnamyl acetate and benzyl benzoate predicted by SwissADME

PARAMETERS	RUTIN
Molecular weight	610.52 g/mol
Number of heavy atoms	43
Number of aromatic heavy atoms	16
Fraction Csp3	0.44
Number of rotatable bonds	6
Number of Hydrogen bond acceptors	16
Number of Hydrogen bond donors	10
Molar refractivity	141.38
Topological polar surface area (TPSA)	269.43 $Å^2$

Table 3.3 ADMET properties of Rutin

PROPERTIES	RUTIN
Water solubility	Soluble
Gastrointestinal (GI) absorption	Low
Skin permeability	-10.26 cm/s
P-glycoprotein substrate	Yes
Lipophilicity (Log Po/w)	1.58
Plasma Protein Binding (PPB)	83.811%
Volume Distribution (VD)	0.754
Total Clearance	1.349
Half-life ($T_{1/2}$)	0.524

Skin sensitization	No
Eye corrosion	No
Respiratory toxicity	No
Acute Toxicity Rule	0 Alert

3.5 Methodology

3.5.1 **Molecular Docking**

1. Protein structure retrieval

In order to perform molecular docking of rutin, structures of proteins involved in the autophagy and metabolism processes were required. These structures were taken from RCSB Protein Data Bank (RCSB PDB) (https://www.rcsb.org/; Berman HM). These structures were downloaded in ".pdb" format, since the subsequent tools to be used for molecular docking supports this particular format of the structure.

The proteins were selected based on their roles in induction of autophagy and metabolism processes in cancer cells respectively. The following table (Table 3.4) is a list of proteins, their roles in the processes and the known-inhibitors of each selected protein which were used to dock in order to compare the binding energy (S) against Rutin:

Table 3.4. List of proteins and inhibitors selected for docking

AUTOPHAGY		
PROTEINS	FUNCTIONS	INHIBITORS
mTORC1 (PDB ID: 8ER6)	Senses fluctuations in extracellular and intracellular nutrients to modulate cellular growth, metabolism and survival	Rapamycin (Sirolimus), PP242, XL388, TORIN1 (Amin, A.G., et al.,2021)
ULK1 (PDB ID:4WNO)	ULK1 forms a complex with ATG13, RB1CC1 (FIP200), and ATG101, integrating signals	SBI-0206965 (Martin, K.R., et al., 2018)

	from mTORC1 and AMPK pathways to initiate early autophagic membrane production.	
METABOLISM		
Phosphofructokinase-1 (PDB ID: 7LW1)	PFK-1 is a rate controlling enzyme in glycolysis, catalyzes the phosphorylation of fructose6-phosphate (F6P) to fructose-1,6-bisphosphate (F1,6BP).	2,5-Anhydroglucitol-1,6-Biphosphate (Lang, L., et al., 2019) and Glucose-3-phosphate (Alshawi, A., & Agius, L.,2019).
Hexokinase- II (PDB ID: 2NZT)	The initial step in glycolysis involves the conversion of glucose to glucose 6-phosphate (G-6-P) via phosphorylation, a process strongly influenced by the enzyme hexokinase II (HK II).	3-BrPA and 2-DG (Liu, B., et al., 2023)
Fatty Acid Synthase (PDB ID:4Z49)	FASN is a key enzyme in lipid metabolism responsible for catalyzing the synthesis of long-chain fatty acids from acetyl-CoA and malonyl-CoA, playing a crucial role in energy storage and cell membrane biosynthesis.	C75 and Cerulenin (Lupu, R., & Menendez, J. A.,2006)
Glutaminase (PDB ID: 8BSK)	Glutaminase, a mitochondrial enzyme, converts glutamine to glutamate and ammonia, playing a crucial role in glutamine metabolism and supporting cancer cell proliferation through glutaminolysis.	Bis-2-(5-phenylacetamido-1,2,4-thiadiazol-2-yl)ethyl sulfide and CB839 (Wu, C., et al., 2018)

2. Quality validation of selected proteins.

PROCHECK, VERIFY3D which is available on UCLA-DOE LAB integrated server (https://saves.mbi.ucla.edu/; Laskowski, RA., et al.,) was used to evaluate the quality of the predicted protein structures. PROCHECK panel generates the Ramachandran plot which is comprised of four different quadrants and visualizes the allowed angles for the protein's backbone. These quadrants are divided into favorable (red), acceptable (yellow), and disallowed (white) regions based on steric hindrance. The system considers each amino acid's properties and bond angles to assign its position on the plot. Protein structures are uploaded one at a time in ".pdb" format. By analysing the Ramachandran plot for each protein, the system can evaluate its overall structure and identify any potential problems. Whereas, VERIFY3D panel takes a protein's 3D structure (provided in a ".pdb" file) and compares it to the protein's amino acid sequence (its 1D structure). By allotting compatibility scores, VERIFY3D examines how well the 3D arrangement aligns with the expected structure for each amino acid type. This includes factors like the location and environment of each amino acid within the protein. High scores indicate a well-fitting structure, while low scores suggest potential errors in the 3D model. This method provides valuable insights into the overall quality and accuracy of a protein's predicted 3D structure, which will be further used to dock with Rutin.

VoroMQA (Voronoi tessellation-based Model Quality Assessment) (https://bioinformatics.lt/wtsam/voromqa;Olechnovič K, Venclovas Č), analyses interatomic contact areas, considering both explicit interactions and solvent effects. It generates comprehensive scores at various levels; global, per residue, and even per atom. It assigns scores (0-1) globally, per residue, and even per atom. High scores throughout the protein structure, especially in favourable Ramachandran plot regions, indicate a good quality model.

3. Ligands

The 3D molecular structure of Rutin was downloaded from PUBCHEM (https://pubchem.ncbi.nlm.nih.gov/) in structure-data format (.sdf) file. Similar steps were followed to download the 3D structures of known-inhibitors of the respective proteins which showed highest binding affinity upon docking.

Rapamycin (or sirolimus, the official generic name) is the prototypical inhibitor of the molecular target of rapamycin (mTOR). The volume of distribution of sirolimus is large (~12 l/kg), indicating wide distribution into tissues and necessitating, in many instances, a loading dose. It was selected to compare the binding energy with mTORC1 against the binding energy of rutin with mTORC1. Similarly, other ATP competitive inhibitors such as TORIN1, PP242 and XL388 were selected.

4. Receptors preparation

The downloaded .pdb structures of proteins that passed the protein validation were opened in AutoDockTools-1.5.7 where all co-crystallized ligand and co-crystallized hetero atoms, including water molecules, were removed. This ensures the docking simulation focuses solely on the protein itself. Hydrogen atoms were added to the protein structure. These play a crucial role in protein-ligand interactions during docking. Gasteiger charges were also calculated for the protein atoms. These charges represent the electron distribution within the molecule and influence how it interacts with other molecules. This was saved in a “. pdbqt” format for further docking processes. A grid box was created around the protein's active site, the region where the Rutin is expected to bind. The size parameters of this grid box were noted down in a separate ".txt" file and was saved for future reference (Morris, G.M., et al)

5. Ligand preparation

The “.sdf” file of Rutin structure was also opened in AutoDockTools-1.5.7 and was converted into “. pdbqt” file.

6. Molecular Docking process

Molecular docking is a computational technique used to analyse the interactions between two or more molecular structures in various conformations. Essentially, docking predicts the best way one molecule can attach to another to form a stable complex. This process evaluates the extent of interaction between the molecules, providing insights into their binding affinities. By predicting the preferred orientation of one molecule to a second when bound to each other to form a stable complex, molecular docking helps in understanding the molecular basis of the interaction. It is widely used in drug discovery and development to predict the binding affinity and activity of small molecules against target proteins. The docking process involves scoring functions that estimate the strength and stability of the

interactions, which is crucial for identifying potential drug candidates and understanding protein-ligand interactions at a detailed level.

AutoDock Vina was selected as the main program to carry out the docking process. The ". pdbqt" files for both the protein and the ligand, along with the size.txt file containing the grid dimensions, were saved in a single file. This file also included the program files of AutoDock Vina, which comprised vina.exe (the executable file), license (the licensing information), and vina_split (a utility for splitting multi-model output files). By organizing all necessary files in one file, the docking process was efficiently managed, allowing AutoDock Vina to predict how Rutin binds to the target proteins (Morris, G.M., et al).

The molecular docking of Rutin with the chosen proteins was performed using AutoDock Vina through the Command Prompt. The command used was:

"vina.exe --receptor receptor.pdbqt --ligand ligand.pdbqt --config size.txt --exhaustiveness 8 --out out.pdbqt"

Here's a breakdown of what this command does:

- vina.exe is the executable file for running AutoDock Vina.
- --receptor receptor.pdbqt specifies the receptor file, which is the protein structure in the .pdbqt format.
- --ligand ligand.pdbqt specifies the ligand file, which is the Rutin molecule also in the .pdbqt format.
- --config size.txt provides the configuration file that contains the grid dimensions defining the active site for docking.
- --exhaustiveness 8 sets the thoroughness of the search; higher values mean more exhaustive searching for the best binding positions.
- --out out.pdbqt specifies the output file where the docking results will be saved.

7. Visualization of the docked complexes

The docked complexes were saved in the ".pdbqt" format for further analysis. To have clear insights regarding the interacting residues of the proteins with Rutin, BIOVIA Discovery Studio Visualizer ver. 2021 was used. This tool was specifically utilized to clearly infer the two-dimensional representations of the binding site atoms (Dassault Systems Biovia)

The three-dimensional visualisation of the docked complexes was done using UCSF Chimera 1.17.3 (Pettersen EF, et al.)

3.5.2 **Nanoparticle Synthesis**

Rutin-loaded nanoparticles were prepared using Solvent-Anti solvent type of nanoprecipitation. Solvent-anti solvent nanoprecipitation is a method used to create nanoparticles by exploiting the solubility differences of a substance in two different liquids. Initially, the drug along with the polymer of interest is dissolved in a solvent where it is highly soluble. This solution is then rapidly introduced into an anti-solvent, a liquid in which the polymer is poorly soluble. As the solvent and anti-solvent mix, the solution experiences a sudden decrease in solubility, leading to supersaturation. Supersaturation causes the substance to aggregate and precipitate out of the solution. Due to the rapid mixing and the high degree of supersaturation, the precipitating substance forms very small particles—nanoparticles. This technique is commonly used because it is straightforward, does not require complex equipment, and allows for the production of nanoparticles with controlled sizes and properties. The size and uniformity of these nanoparticles can be influenced by various factors such as the concentration of the polymer in the solvent, the rate of mixing, the ratio of solvent to anti solvent, solvent removal, anti-solvent concentration and the overall conditions such as temperature, storage etc (Govender, T., et al., 1999; Bilati, U., et al., 2005; Vu, H. T., et al., 2019)

The solvent solution consisted of various concentrations of PEG-PLGA polymer were prepared along with Rutin and dissolved in suitable solvents (DMSO or DCM). The various formulations are shown in the Table 3.5**.** The anti-solvent solution was made up of Polyvinyl alcohol (PVA) dissolved in HPLC grade water. PVA is a well-known hydrophilic polymer and usually acts as stabilizer. Although PVA has hydrophilic –OH group, it has also a nonpolar vinyl part. Thus, it reduces the surface tension between the aqueous part and the nonpolar non-aqueous part, where vinyl group remains towards non-polar part and OH-group faces towards aqueous part. Thus, it stabilizes primary emulsion. Previously prepared PVA solution in HPLC grade water was kept on the magnetic stirrer to stir continuously and the solvent was added drop-wise using a syringe pump through various controlled flowrates. Then the nanosuspension was subjected to overnight spinning on the magnetic stirrer to facilitate the solvent removal. The suspension was then subjected to centrifugation and was

washed with HPLC grade water, to separate the nanoparticles. The pellets were collected after discarding the supernatant. Furthermore, the pellets were subjected to drying in order to get a fine powder of nanoparticles. Various batches of nanoparticles were prepared using the above method but without the drug in order to optimize the nanoparticles size (Almoustafa, H. A., et al., 2017)

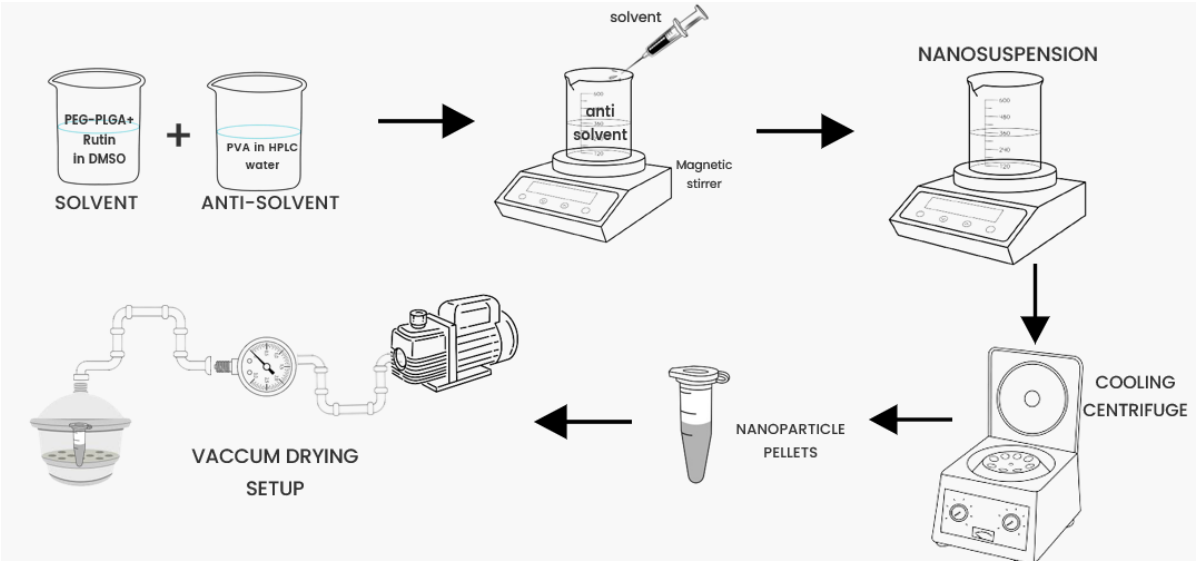


Fig 3.4 Nanoparticles synthesis by solvent: anti-solvent method

- Surface morphology by field emission scanning electron microscopy (FESEM)

Surface morphology of the nanoparticles was analysed using field emission scanning electron microscope (Model-JSM-7100F; Centre for Nano and Material Sciences (CNMS), JAIN global Campus and Apreo 2 SEM; ThermoFisher Scientific; Christ University). Autofine coater was used (either Gold-sputtering or Chromium-sputtering) to enhance the surface conductivity of the material. The coating was done at 10mA for 150 seconds to 300seconds. Then the coated samples were loaded on brass studs and a conductive adhesive tape was used to cover the sample and the studs were then placed inside the stud holder of the FSEM machine and the nanoparticles' structure was visualized in order to determine the size of the particles.

Table 3.5. The various formulations done for nanoparticle synthesis

Trial No.	Solvent used	PEG–PLGA conc.	PVA conc	Anti-Solvent volume	Solvent to Anti-Solvent ratio (SAS)	Magnetic stirring speed	Syringe pump flow rate	Centrifugation	Nanosuspension spinning	Drying method used
1	DMSO	20mg/ml (10mg in 0.5ml)	1.5% (75mg in 5ml)	5ml HPLC water		500RPM	0.2ml/min	15000RPM for 1hr (3cycles)	overnight spinning at 500RPM	Lyophilized
2	DCM	10mg/ml (20mg in 2ml	1% (50mg in 5ml)	5ml HPLC water	1:2.5	500RPM	1ml/min	15000RPM for 1hr (3cycles)	overnight spinning at 500RPM	Vacuum drying for 3.5hrs
3	DCM	5mg/ml (10mg in 2ml)	2% (200mg in 10ml)	10ml HPLC water	1:5	800RPM	1ml/min	Ultrasonication (3mins) and 10000RPM for 15mins (3 cycles)	500RPM for 6hrs	Vacuum drying for 4 hrs
4	DCM	5mg/ml (25mg in 5ml)	2% (0.4g in 20ml HPLC water)	20ml HPLC water	1:4	600RPM	1ml/min	10,000 RPM for 15mins (3 cycles)	600RPM for 6hrs	Vacuum drying for 4.5hrs

CHAPTER 4: RESULTS

4.1 **Molecular docking results**

1. Quality validation of selected proteins.

The quality of the selected proteins was evaluated by Ramachandran plot, available on PROCHECK, VERIFY3D and VoroMQA.

The Ramachandran Plot of mTORC1 protein (PBD ID: 8er6) suggested that 93.2% of the amino acid residues lied in the favorable region while the rest of the residues were allocated in additionally allowed and generously allowed regions. No part of the residues was allotted in disallowed portion of the plot.

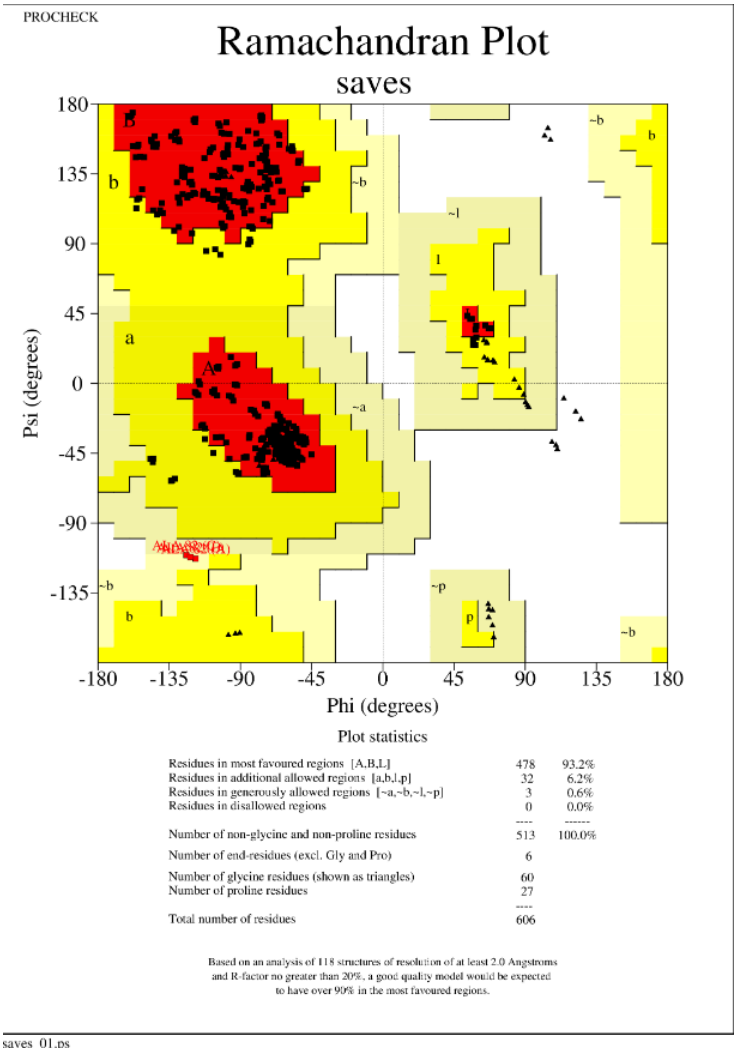


Fig 4.1 PROCHECK result of mTORC1 protein (PDB ID: 8ER6)

Over 98.51% of the residues have averaged 3D-1D score >= 0.1 in the VERIFY3D server indicating a well-fitting 3D structure.

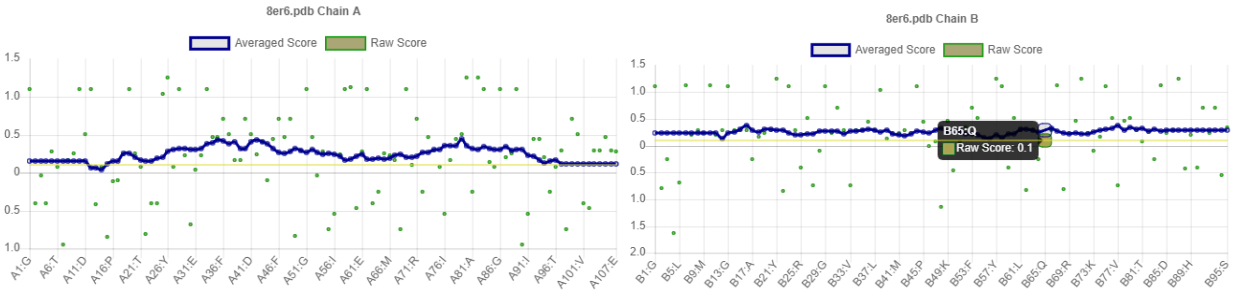


Fig 4.2 VERIFY 3D results of 8ER6

VoroMQA analysis of mTORC1 gave a score of 0.486 indicating a good quality model.

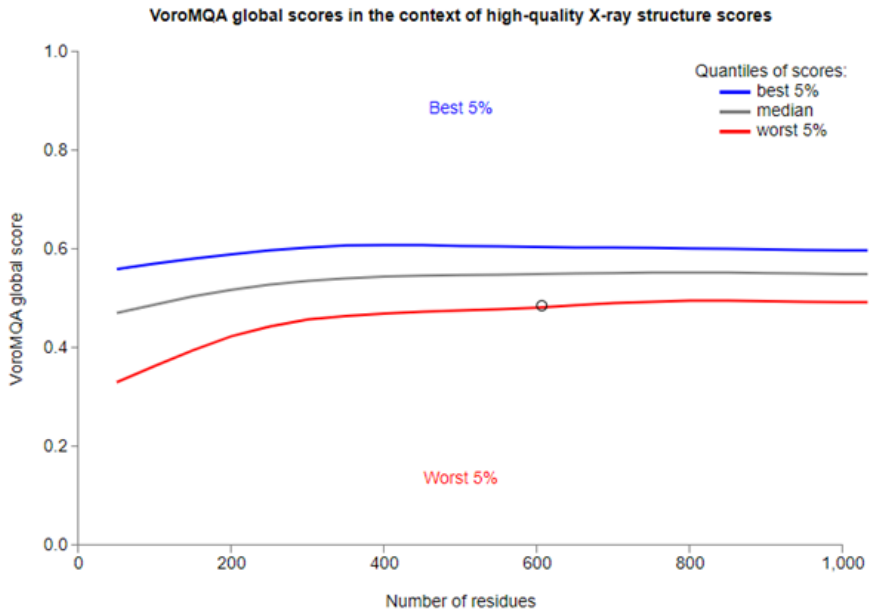
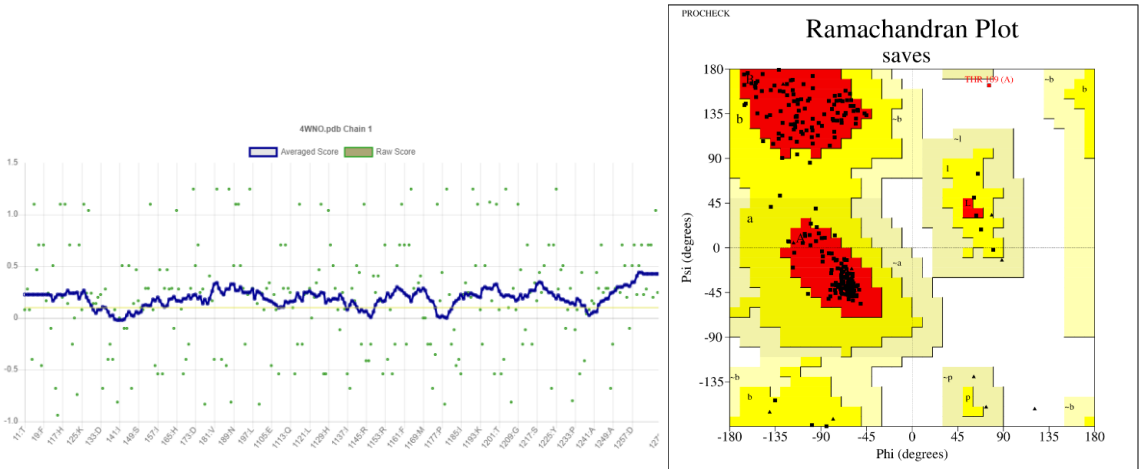
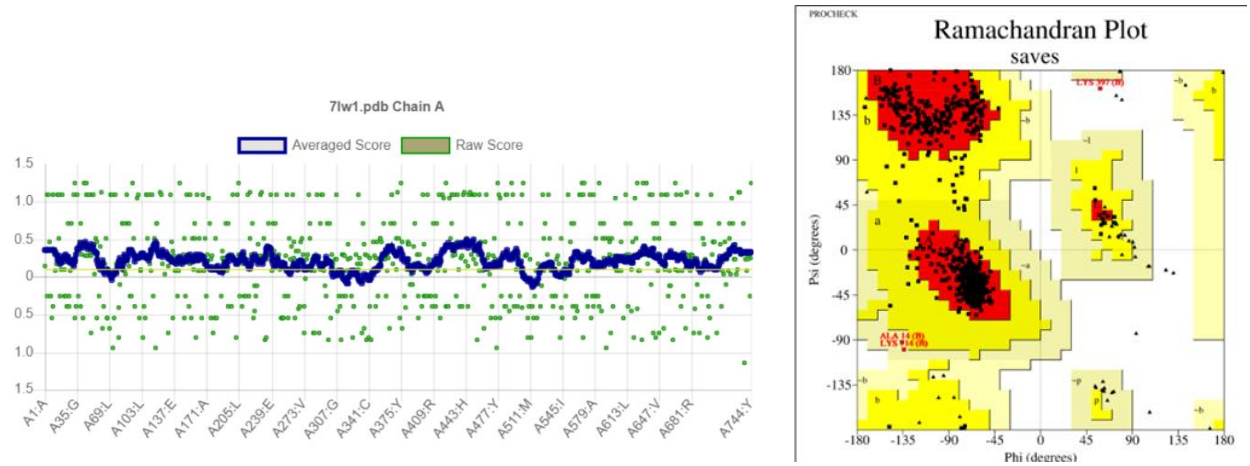


Fig 4.3 VoroMQA Global Score Plot of 8ER6

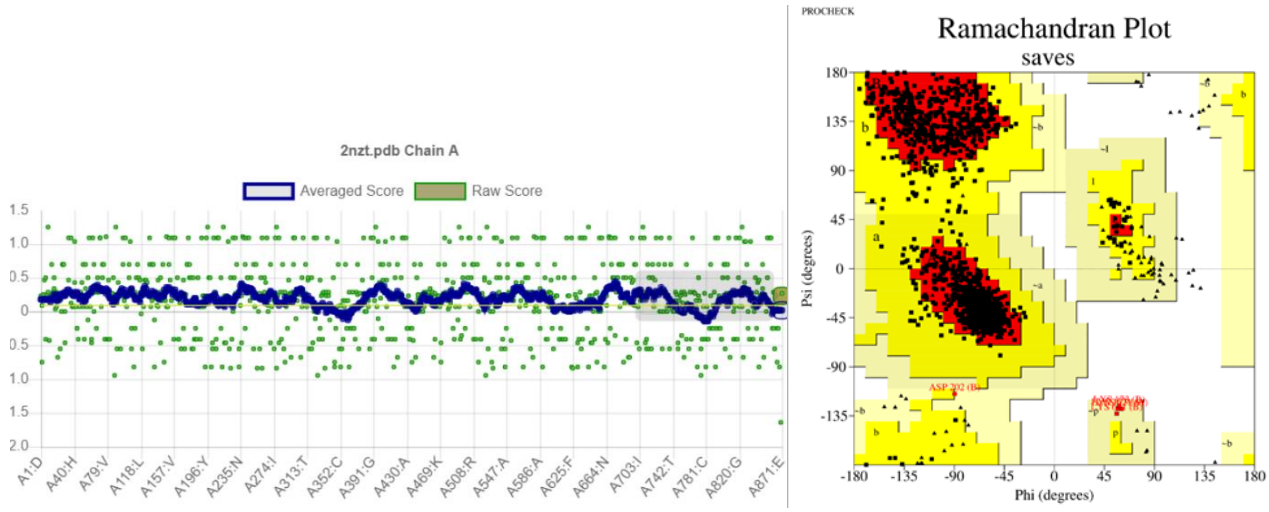
A similar screening was done for all the selected proteins.


4WNO.pdb Chain 1
Averaged Score
Raw Score
PROCHECK
Ramachandran Plot
saves
Psi (degrees)
Phi (degrees)

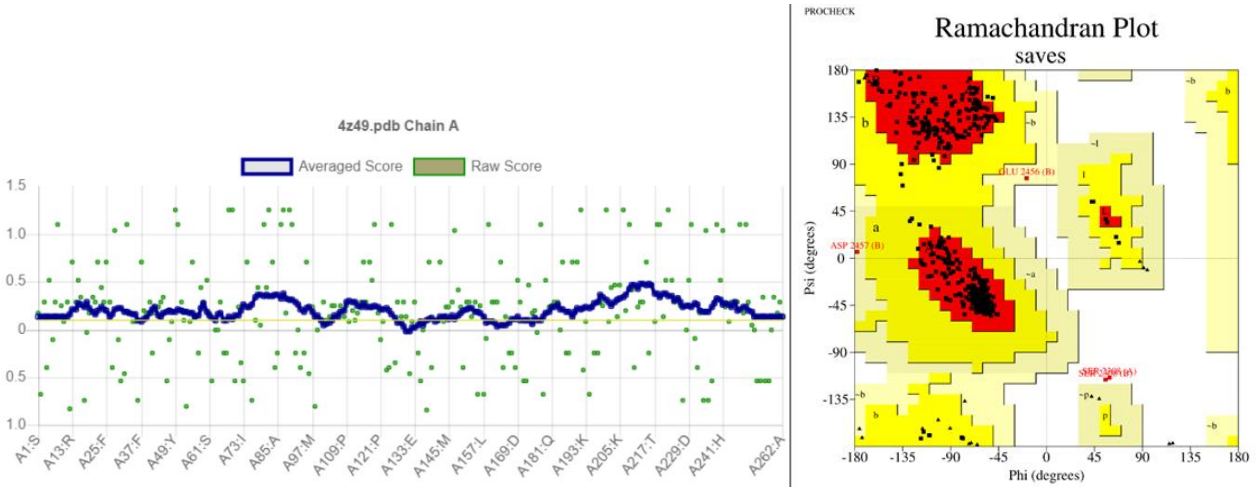
a)


7lw1.pdb Chain A
Averaged Score
Raw Score
PROCHECK
Ramachandran Plot
saves
Psi (degrees)
Phi (degrees)

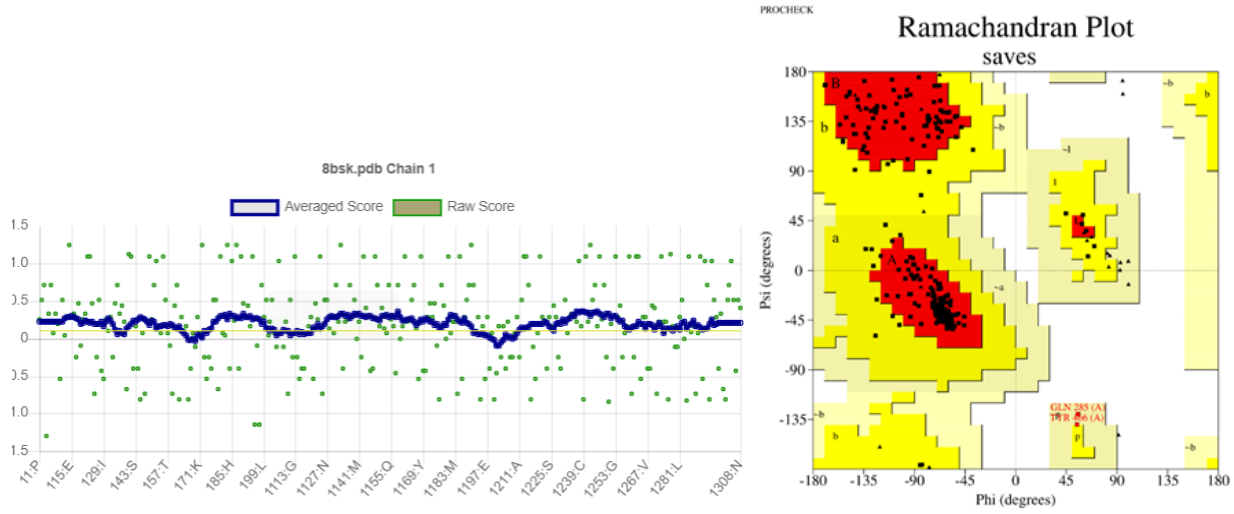
b)


2nzt.pdb Chain A
Averaged Score
Raw Score
PROCHECK
Ramachandran Plot
saves
Psi (degrees)
Phi (degrees)

c)



d)



e)

Fig 4.4 PROCHECK AND VERIFY3D results of all the selected proteins; a) ULK1 (PDB ID:4WNO); b) PFK1 (PDB ID:7LW1); c) HK-II (PDB ID:2NZT); d) FASN (PDB ID:4Z49); e) GLS (PDB ID:8BSK).

2. Molecular Docking and virtual screening

The proteins were then incorporated for molecular docking analysis to interact with Rutin.

During the analysis of docking results, we observed that evaluating the binding energy and root mean square deviation (RMSD) is crucial for understanding the stability and affinity of the interactions between molecules. A low binding energy indicates a strong interaction, while a low RMSD signifies a stable binding conformation. Among the various docked

complexes and poses, the one with the best binding energy often fits well within the binding site of the target protein. Additionally, identifying the key interacting residues, i.e., amino acids of the binding site help us understand the specific interactions that contribute to the binding affinity and binding pocket. These insights are essential for determining the potential effectiveness of a compound as a binder to the target protein and helps us compare the binding sites of Rutin and other known inhibitors.

Table 4.1 The binding energy (S) in kcal/mol of mTORC1 with Rutin and all the selected inhibitors.

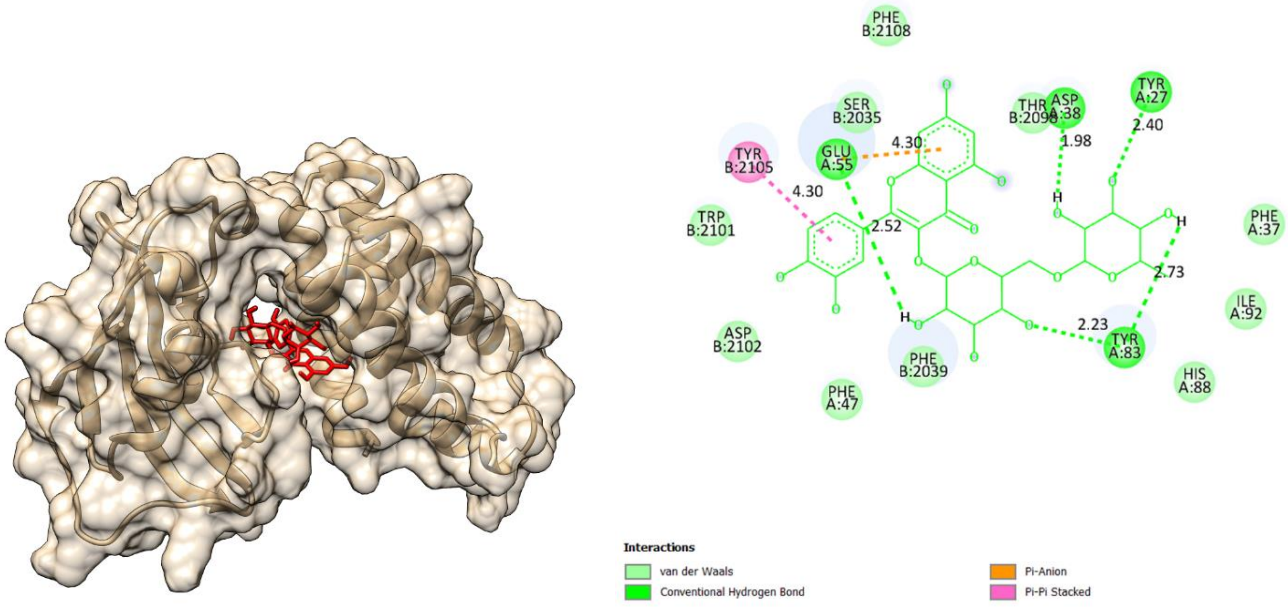
	Rutin	Rapamycin	PP242	XL388
mTORC1	-10.6 kcal/mol	-10.00 kcal/mol	-8.8 kcal/mol	-10.7 kcal/mol

After observing a high binding affinity between mTORC1 and rutin, we proceeded to dock mTORC1 with ATP to determine if rutin acts as an ATP-competitive inhibitor. The docking results revealed that rutin occupies the same binding site as ATP. This overlap in binding sites confirms that rutin competes directly with ATP for binding to mTORC1, thereby establishing its role as an ATP-competitive inhibitor. This finding is significant as it suggests that rutin can effectively inhibit the mTORC1 protein by blocking ATP binding, which is essential for the protein's activity. Understanding this competitive inhibition mechanism provides valuable insights for the development of potential therapeutic interventions targeting mTORC1.

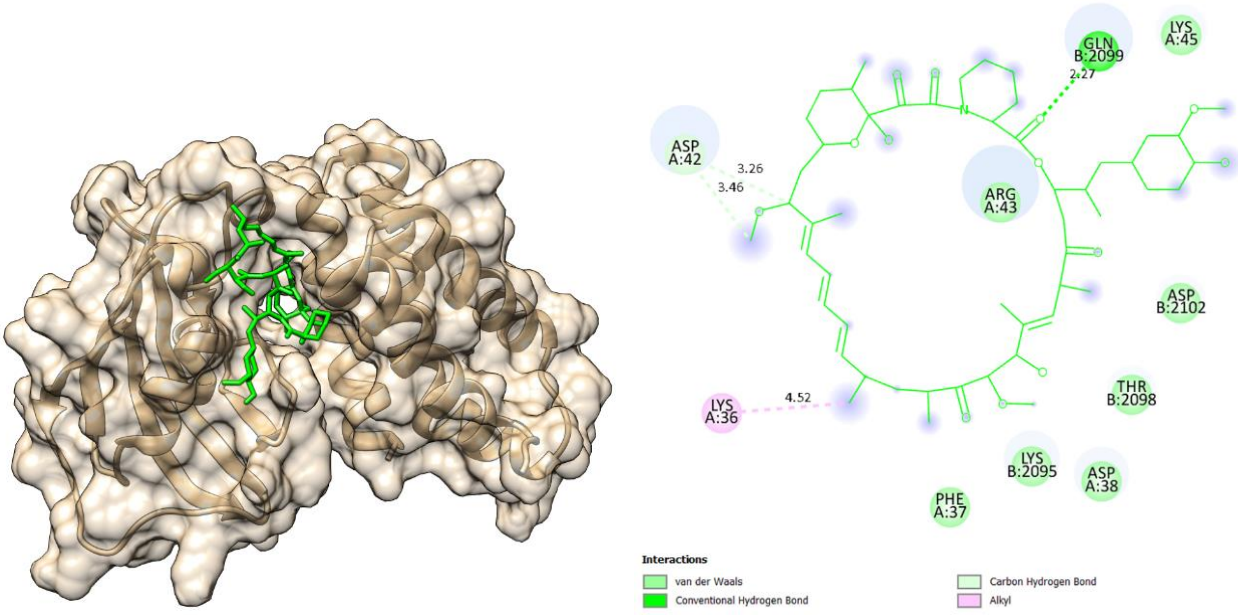
To further investigate the synergistic effects of binding rutin and other known mTORC1 inhibitors—namely Rapamycin, PP242, and XL388—we conducted docking studies using the Multi-Ligand Simultaneous Docking (MLSD) approach. Firstly, the binding energy values indicated increased affinity when rutin was docked with these inhibitors. Notably, rapamycin, a well-known mTORC1 inhibitor, did not bind to the same pocket as rutin, and there was no inter-ligand interaction observed between rutin and rapamycin. This suggests that rutin maintains its binding pocket, which could potentially make it a more effective inhibitor compared to rapamycin alone.

The MLSD score for the rutin and rapamycin complex was -14.336 kcal/mol, indicating a strong binding interaction. In comparison, PP242 and XL388 had MLSD scores of -13.558 kcal/mol and -14.389 kcal/mol, respectively. These scores reflect the binding affinity and potential stability of the complexes. Importantly, PP242 and XL388 showed inter-ligand interactions with rutin, suggesting a synergistic effect that could enhance their inhibitory potential.

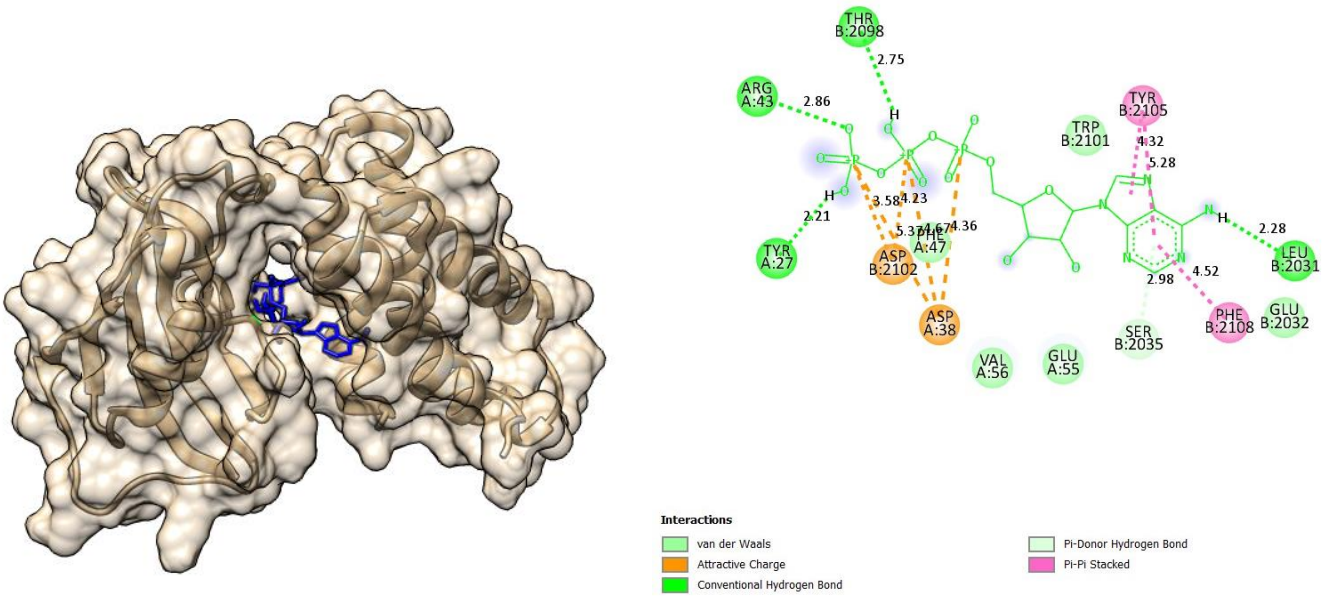
Furthermore, only complexes that did not display unfavorable bonds in their 2D diagrams were selected for detailed analysis. This careful selection ensures the reliability of the docking results and the potential effectiveness of the inhibitor combinations. The observation of synergistic interactions between rutin and both PP242 and XL388 highlights the possibility of improved therapeutic strategies by combining these compounds with rutin.



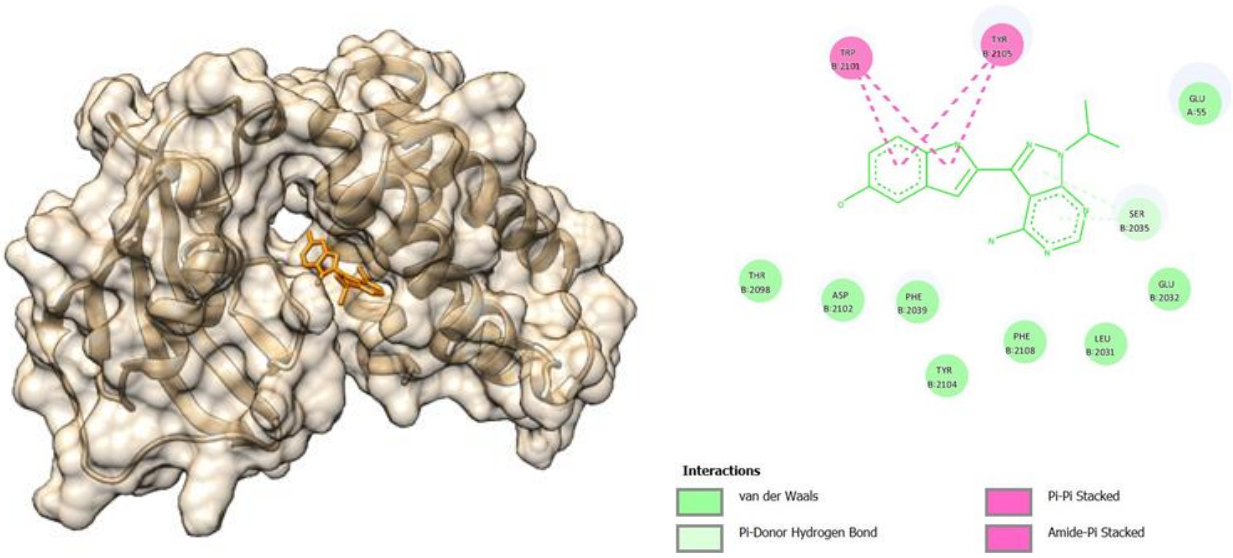
a)


GLN B:2099
LYS A:45
2.27
ASP A:42
3.26
3.46
ARG A:43
ASP B:2102
THR B:2098
4.52
LYS A:36
LYS B:2095
ASP A:38
PHE A:37
Interactions
van der Waals
Conventional Hydrogen Bond
Carbon Hydrogen Bond
Alkyl

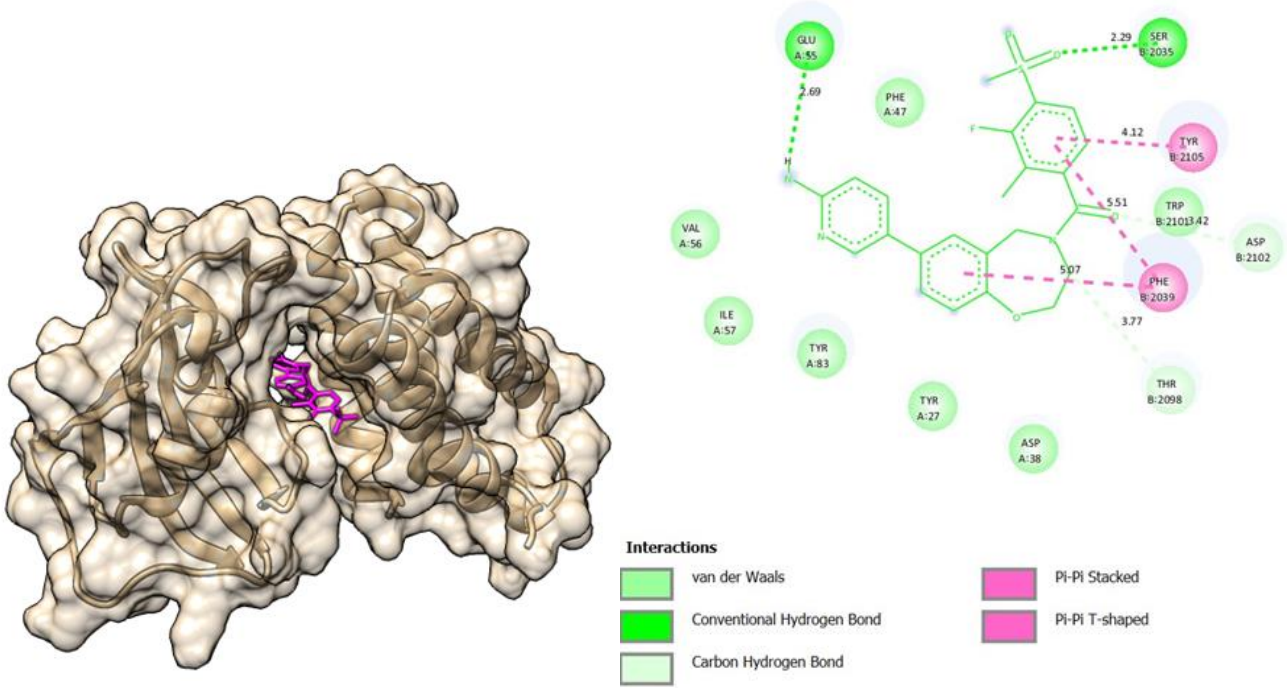
b)


THR B:2098
2.75
ARG A:43
2.86
TYR B:2105
TRP B:2101
4.32
5.28
2.21
3.58
4.23
4.67
4.36
TYR A:27
PHE A:47
ASP B:2102
ASP A:38
2.28
LEU B:2031
2.98
4.52
PHE B:2108
GLU B:2032
SER B:2035
VAL A:56
GLU A:55
Interactions
van der Waals
Attractive Charge
Conventional Hydrogen Bond
Pi-Donor Hydrogen Bond
Pi-Pi Stacked

c)



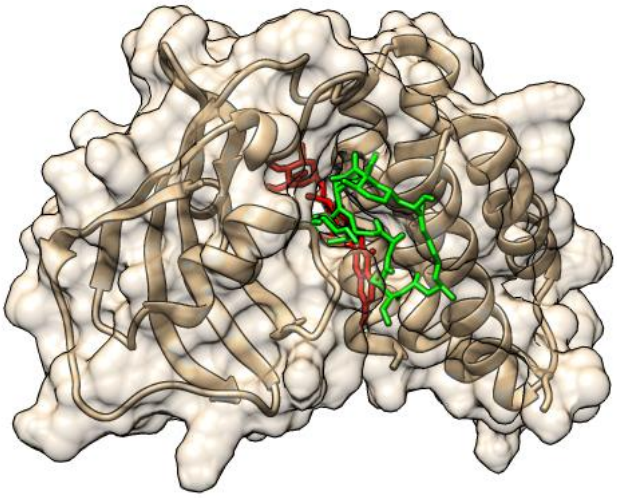
d)



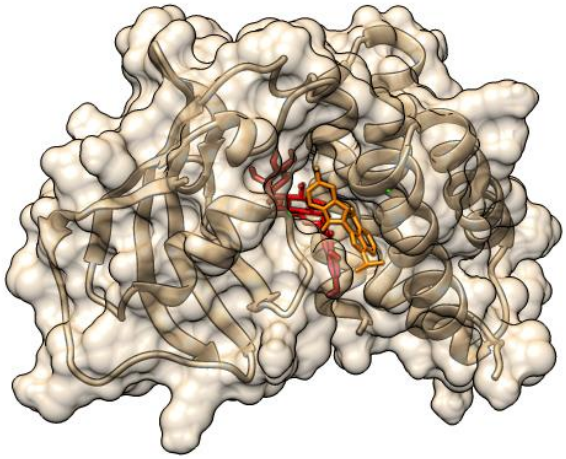
e)

Fig 4.5 Molecular Docking results of mTORC1 with - a) Rutin b) Rapamycin; c) ATP; d) PP242; e) XL388.

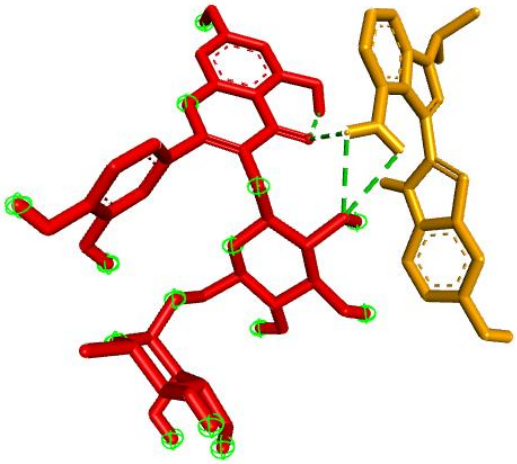
MLSD RESULTS:



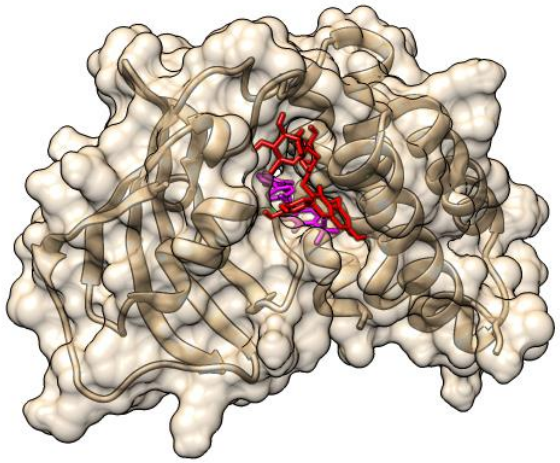
a)



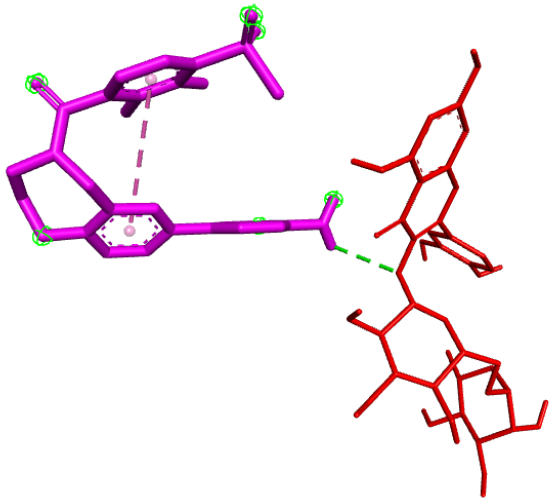
b)



c)



d)



e)

Fig 4.6 Multi-Ligand Simultaneous Docking results of mTORC1 with –

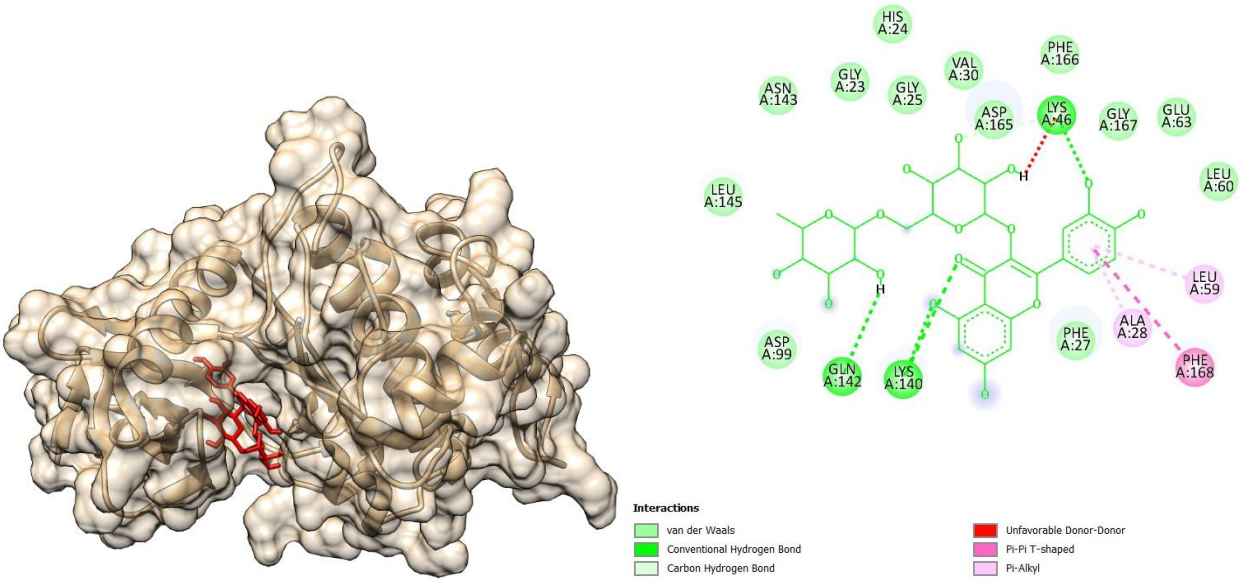
a) Rutin and Rapamycin (S = -14.336 kcal/mol);

b) Rutin and PP242 (S = -13.558 kcal/mol); c) Inter-Ligand interactions between Rutin and PP242;

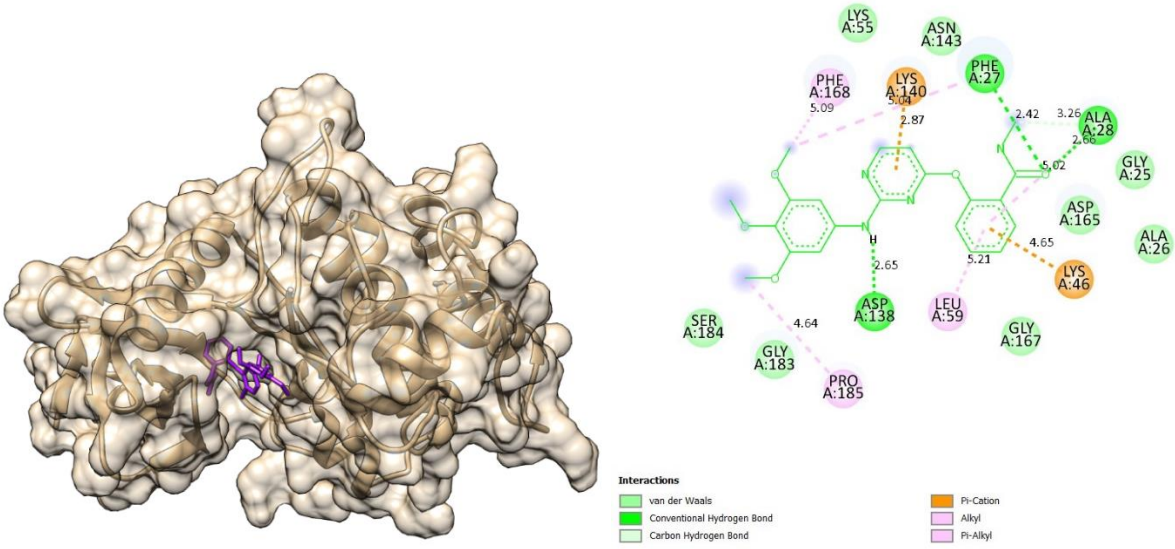
d) Rutin and XL388 (S = -14.389 kcal/mol); e) Inter-Ligand interactions between Rutin and XL388

Table 4.2 The binding energy (S) in kcal/mol of ULK1 with Rutin and it's the selected inhibitors.

	Rutin	SBI-020696
ULK1	-9.5 kcal/mol	-7.4 kcal/mol



a)

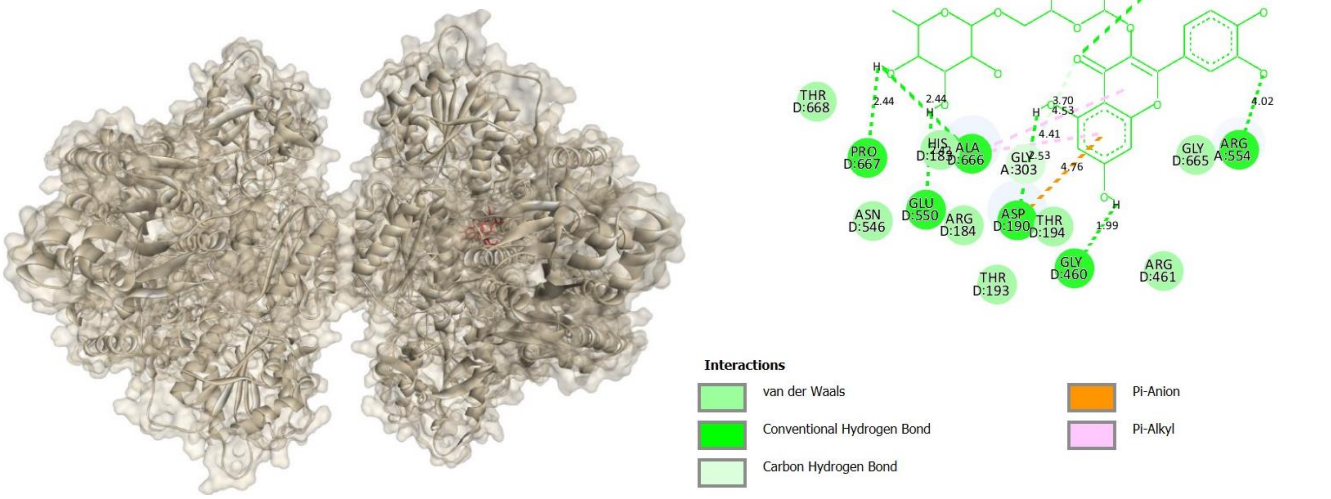


b)

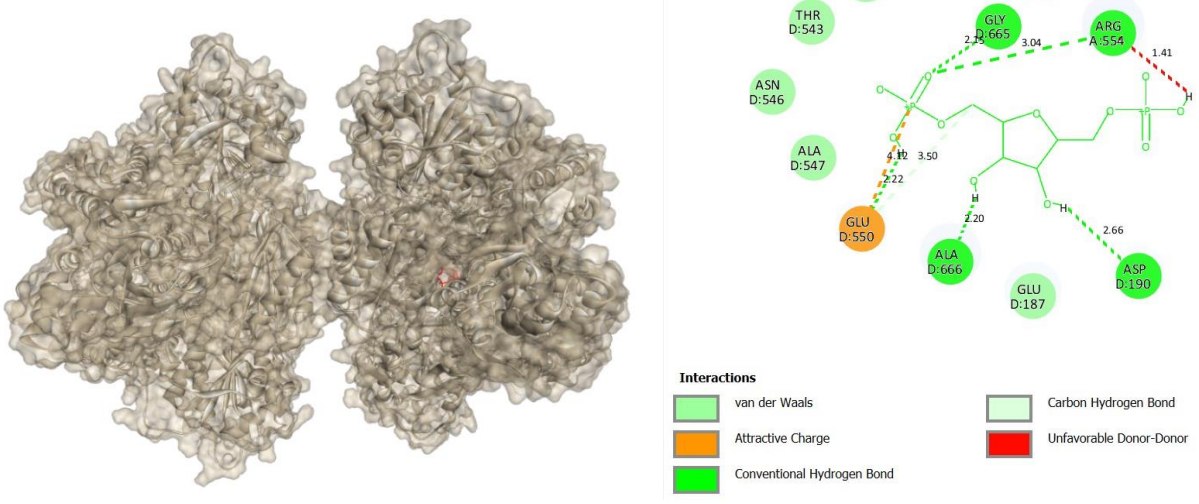
Fig 4.7 Molecular Docking results of ULK1 with - a) Rutin b) SBI-020696

Table 4.3 The binding energy (S) of PFK-1 in kcal/mol with Rutin and it's the selected inhibitors.

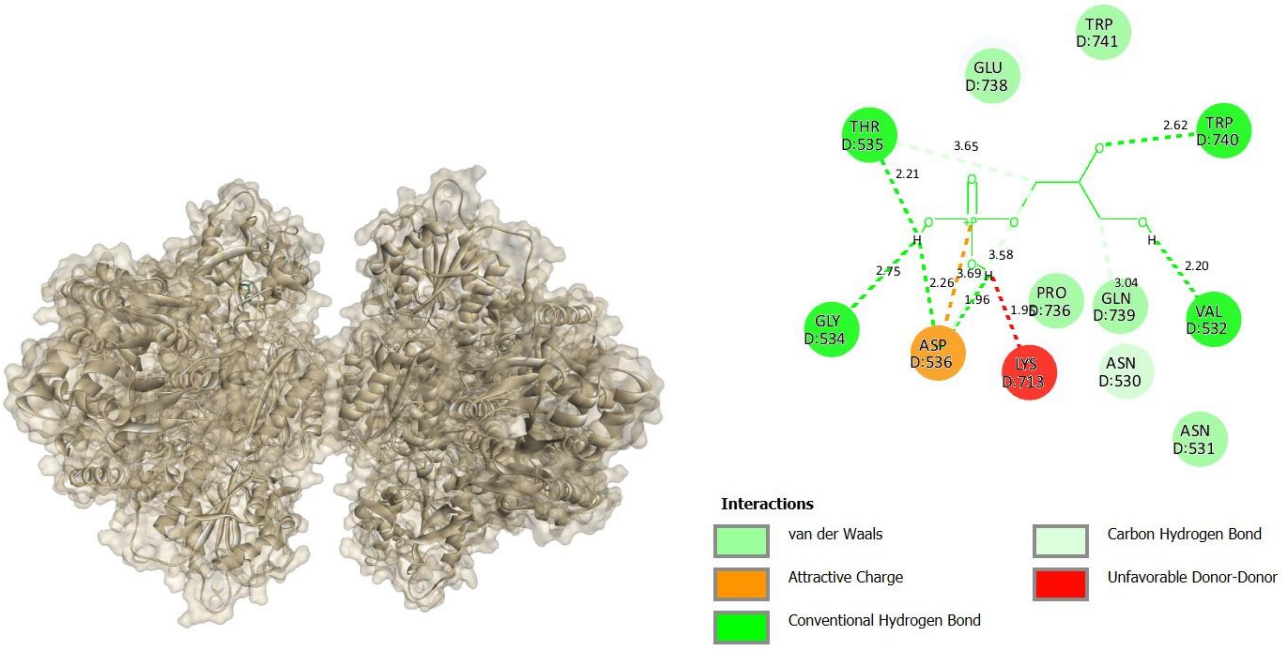
	Rutin	2,5-Anhydroglucitol-1,6-Biphosphate	Glucose-3-phosphate
PFK-1	-9.8 kcal/mol	-5.7 kcal/mol	-4.7 kcal/mol



a)



b)

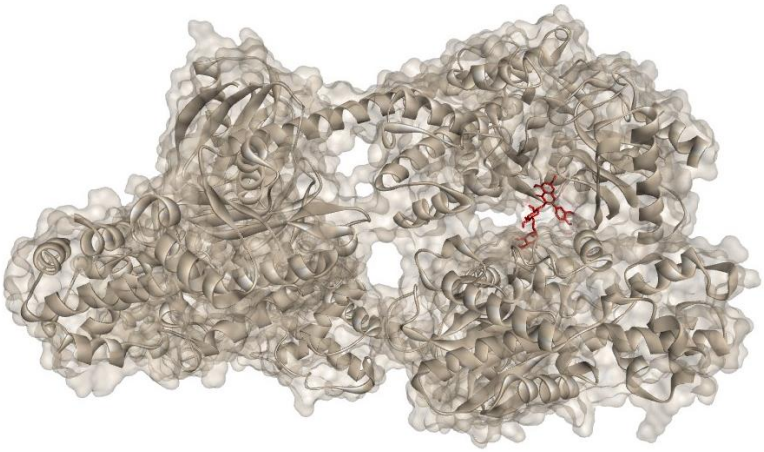


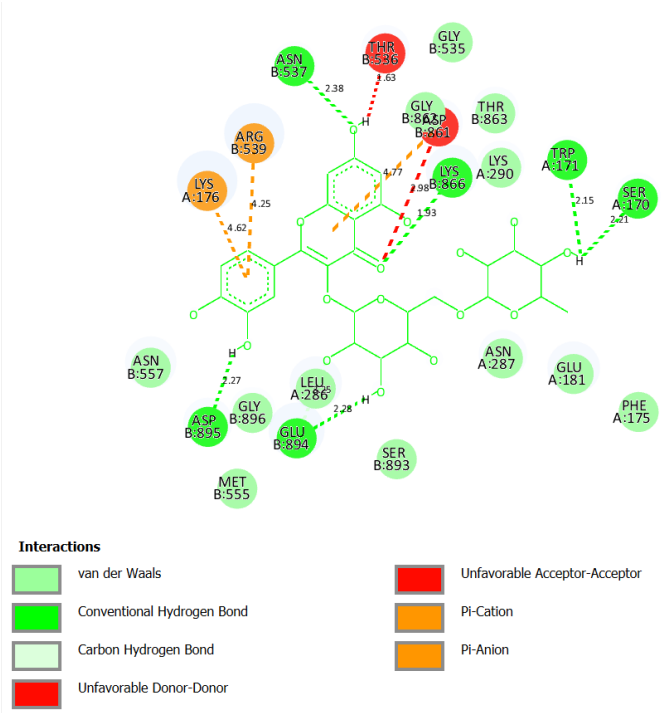
c)

Fig 4.8 Molecular Docking results of PFK1 with - a) Rutin; b) 2,5-Anhydroglucitol-1,6-Biphosphate; c) Glucose-3-phosphate

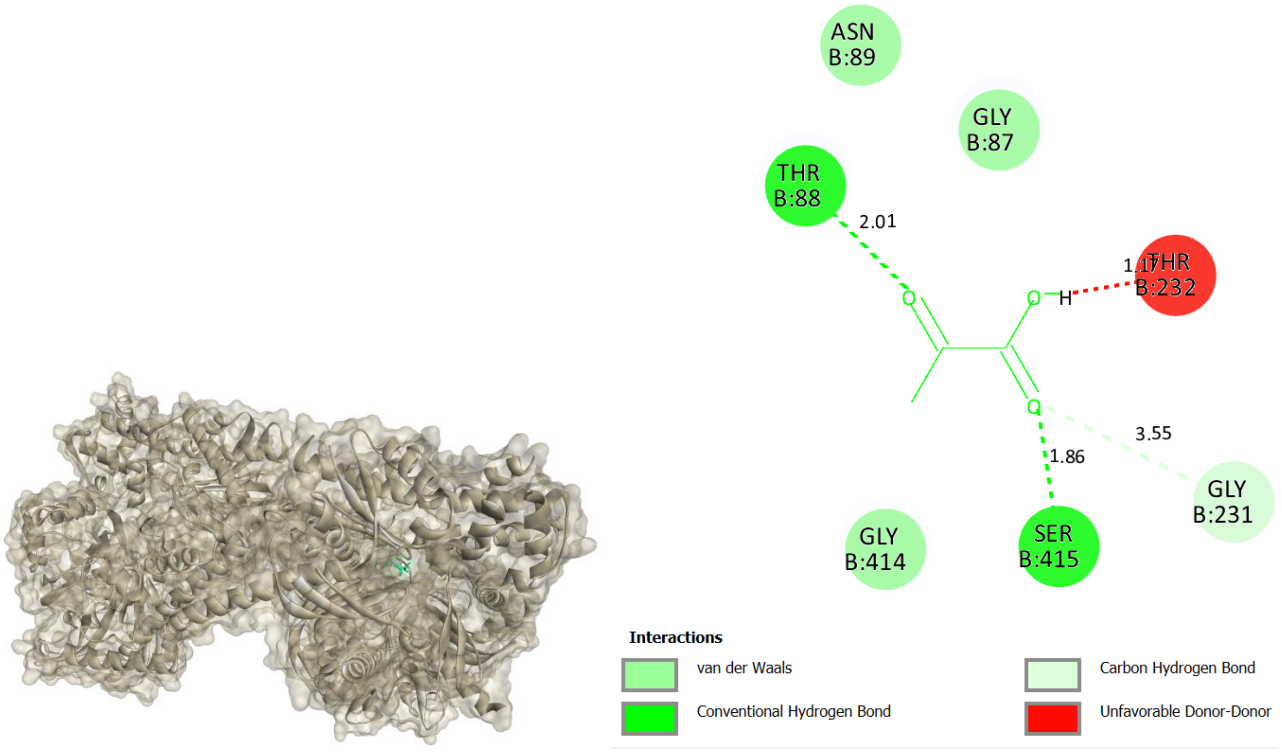
Table 4.4 The binding energy (S) of Hexokinase- II in kcal/mol of with Rutin and it's the selected inhibitors.

	Rutin	3-BrPA	2-DG
HK-II	-9.5 kcal/mol	-5.7 kcal/mol	-5.3 kcal/mol

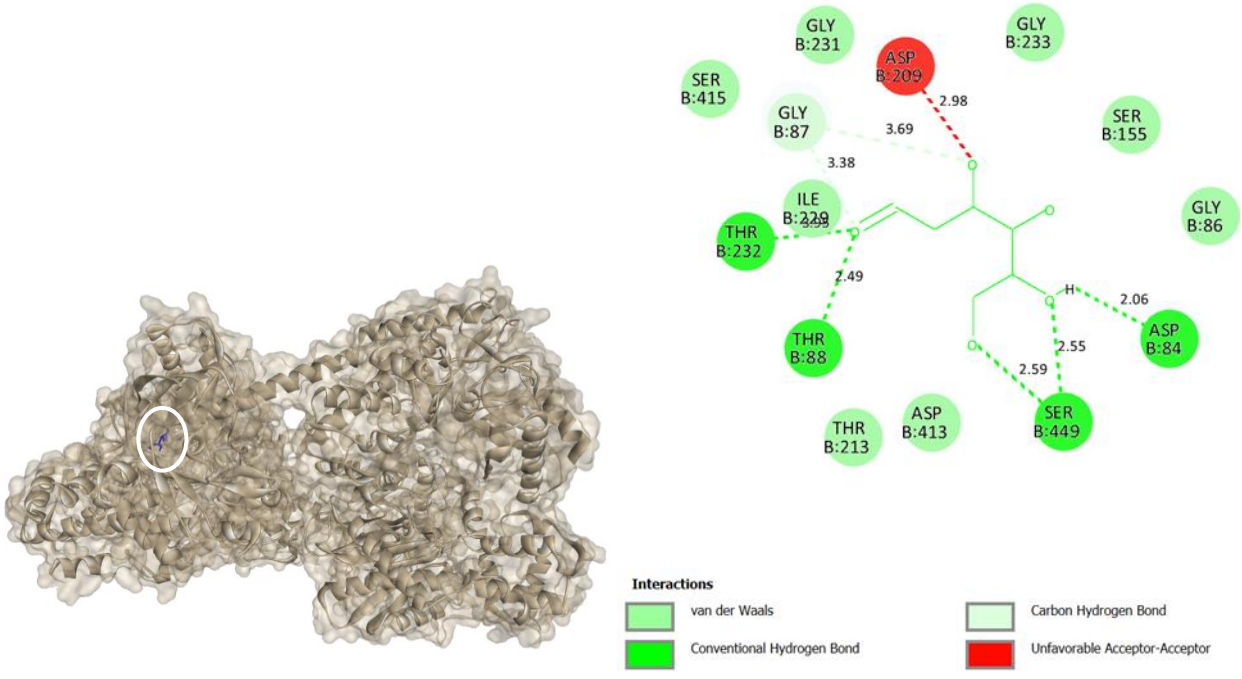




a)



b)

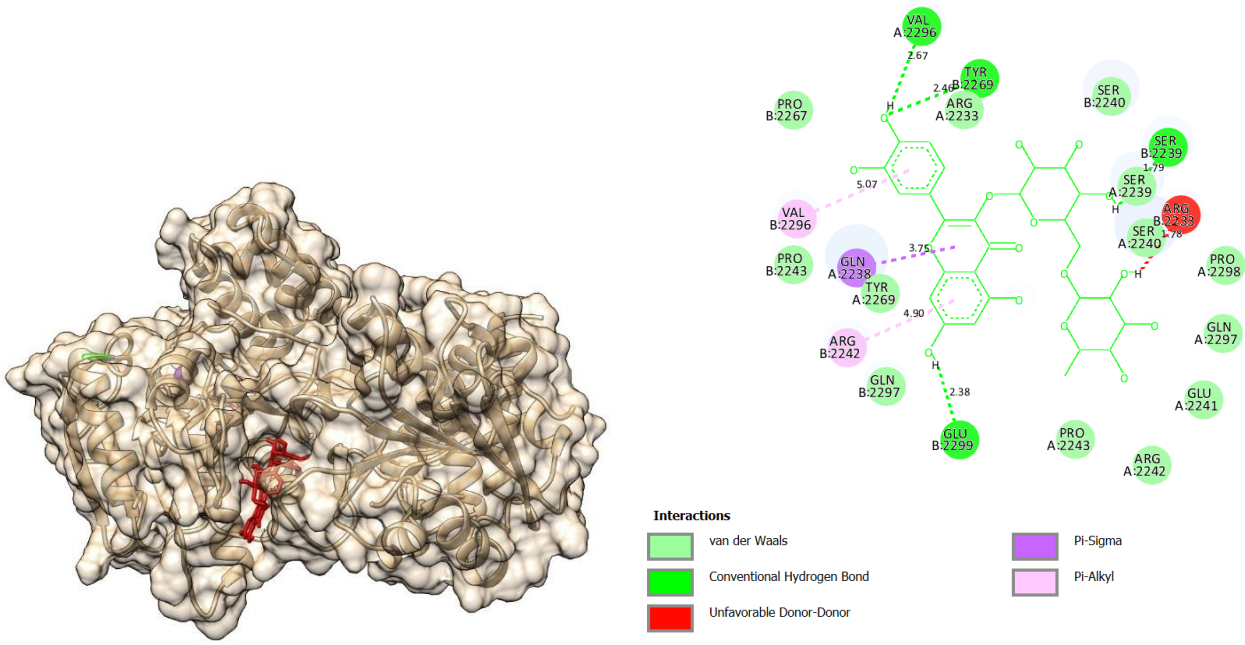


c)

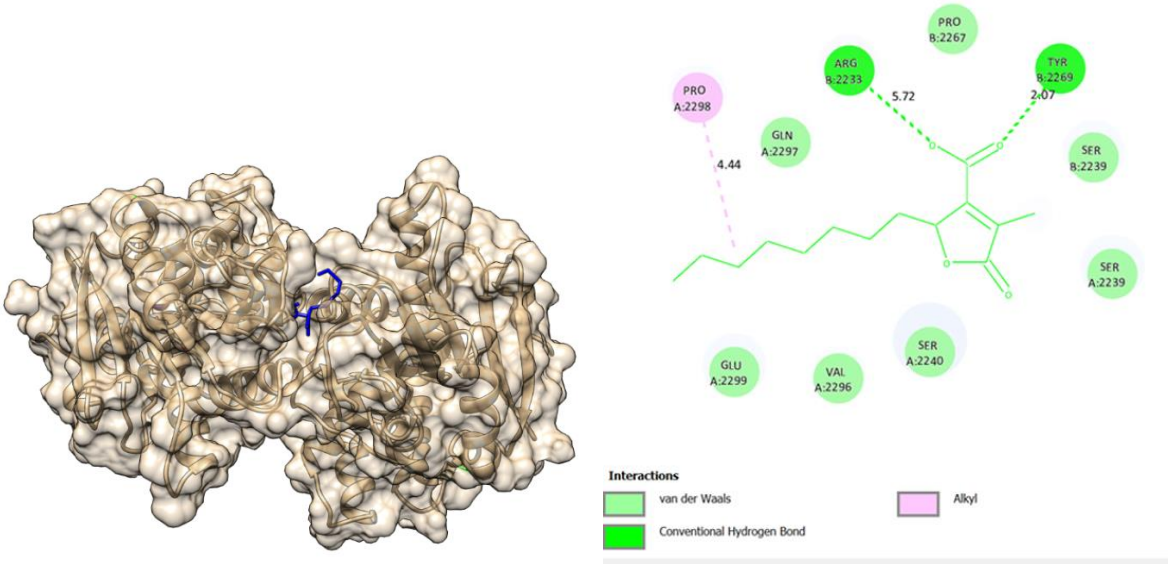
Fig 4.9 Molecular Docking results of HK-II with - a) Rutin; b) 3-bromopyruvate (3-BrPA); c) 2-Deoxy-D-glucose (2-DG)

Table 4.5 The binding energy (S) of Fatty Acid Synthase in kcal/mol of with Rutin and it's the selected inhibitors.

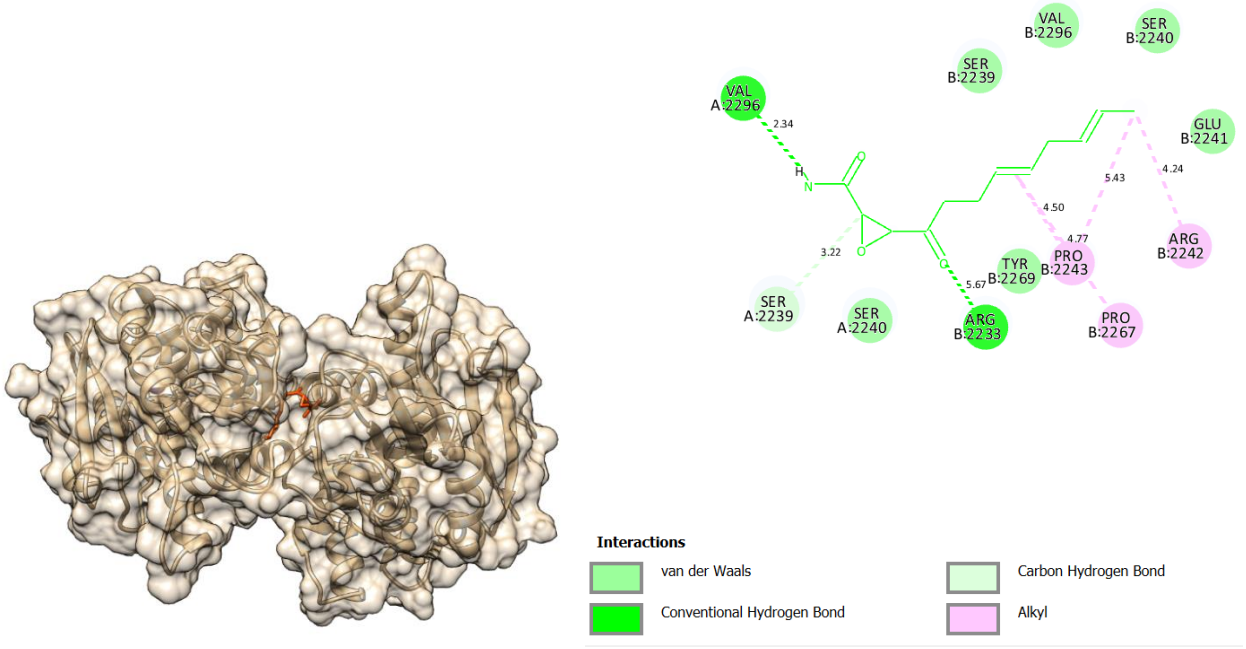
	Rutin	C75	Cerulenin
FASN	-9.6 kcal/mol	-5.4 kcal/mol	-5.5 kcal/mol



a)



b)

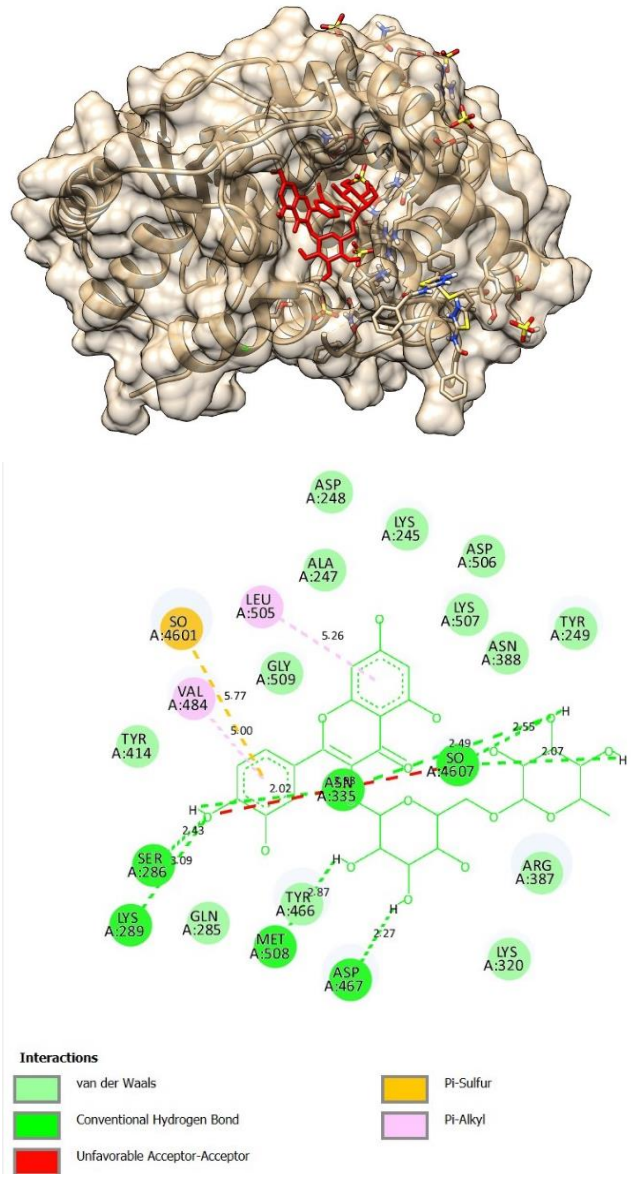


c)

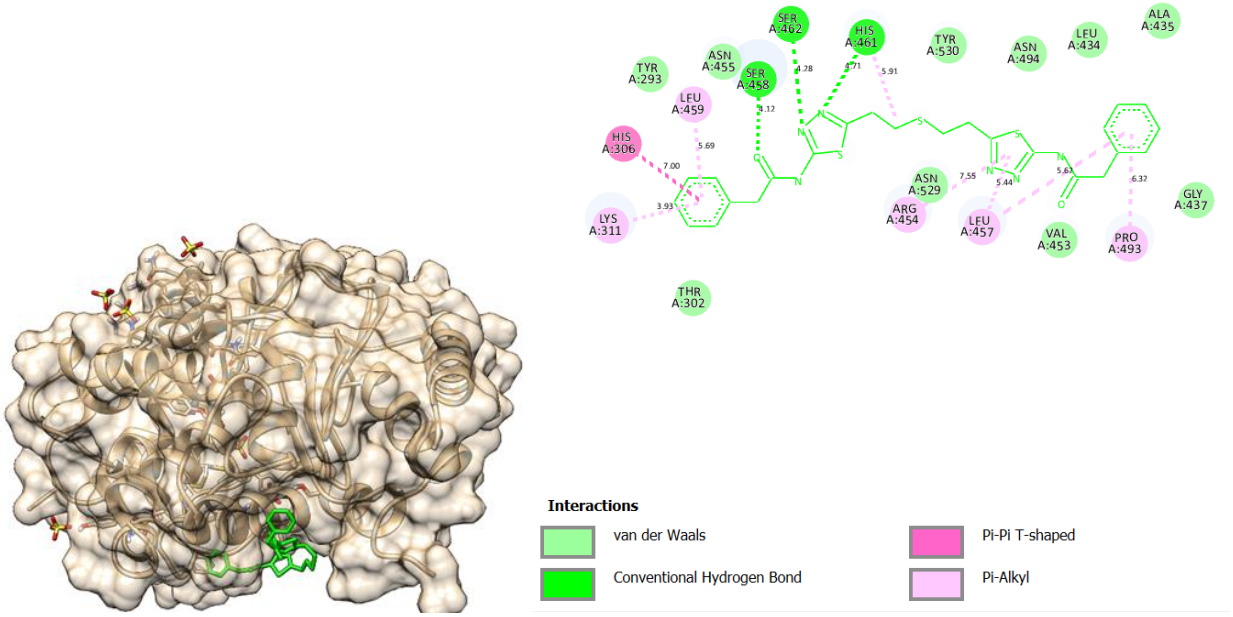
Fig 4.10 Molecular Docking results of FASN with - a) Rutin; b) C75 (4-Methylene-2-octyl-5-oxotetrahydrofuran-3-carboxylic acid); c) Cerulenin

Table 4.6 The binding energy (S) of Glutaminase in kcal/mol of with Rutin and it's the selected inhibitors.

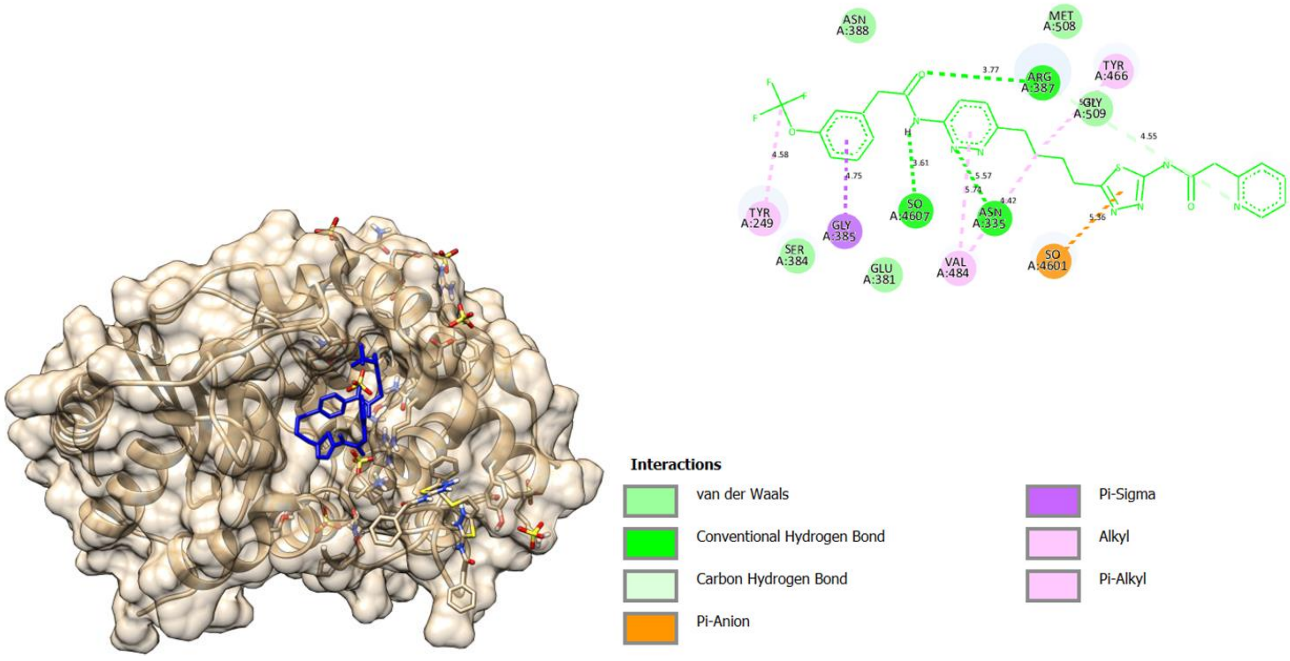
	Rutin	BPTES	CB839
GLS	-8.9 kcal/mol	-5.3 kcal/mol	-7.2 kcal/mol



a)



b)



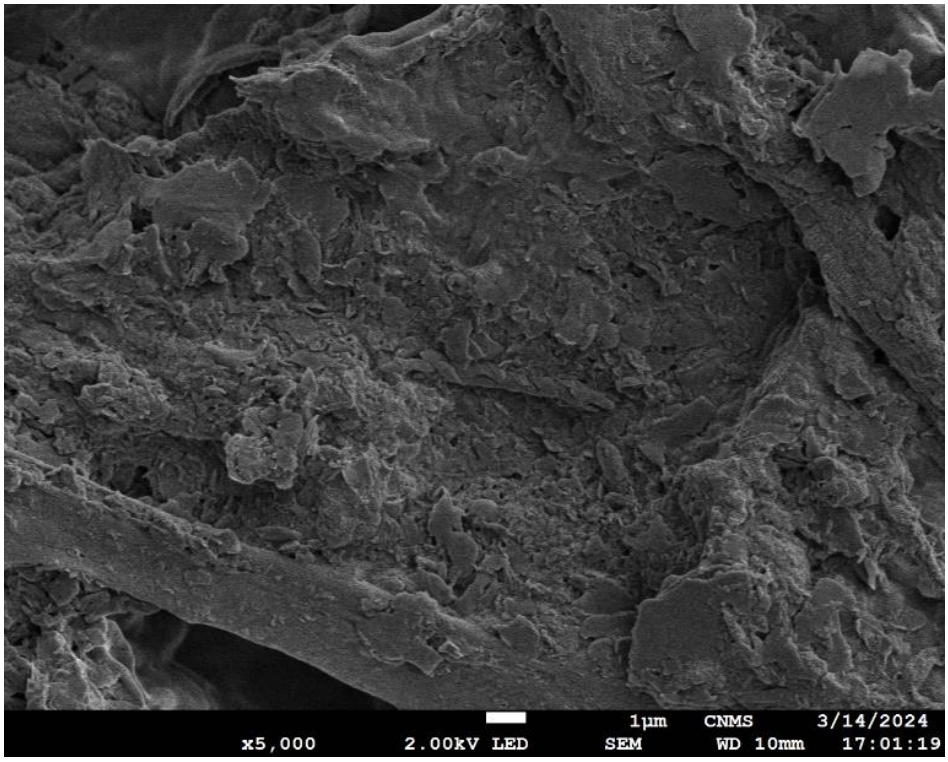
c)

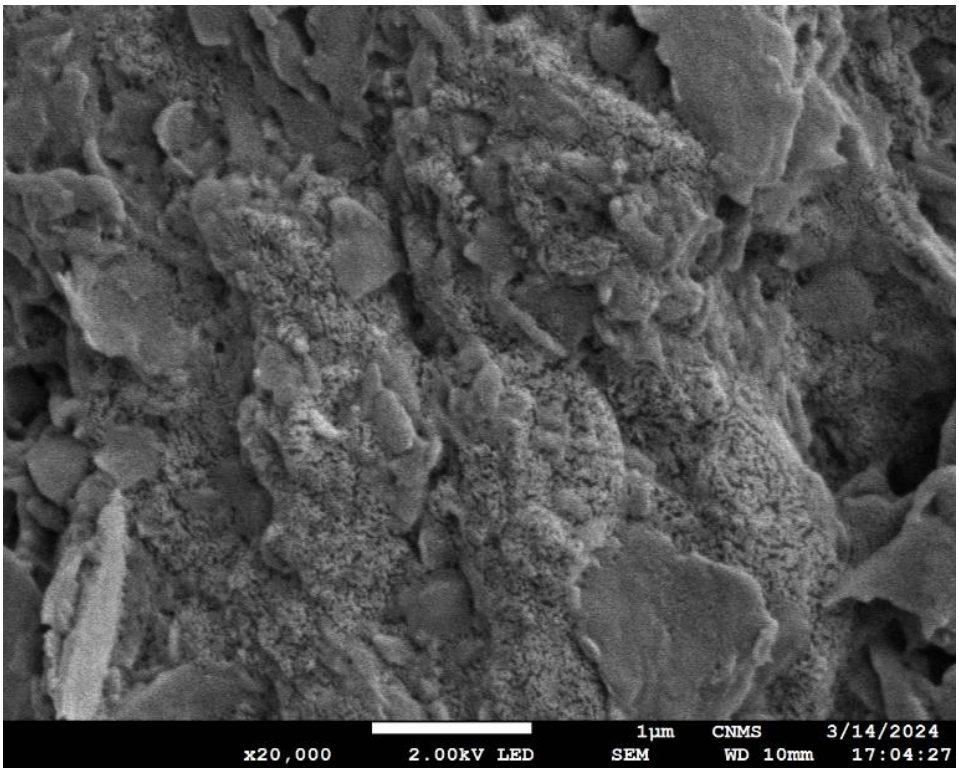
Fig 4.11 Molecular Docking results of GLS with - a) Rutin; b) BPTES); c) CB839 (Telaglenastat).

4.2 **Nanoparticle Synthesis**

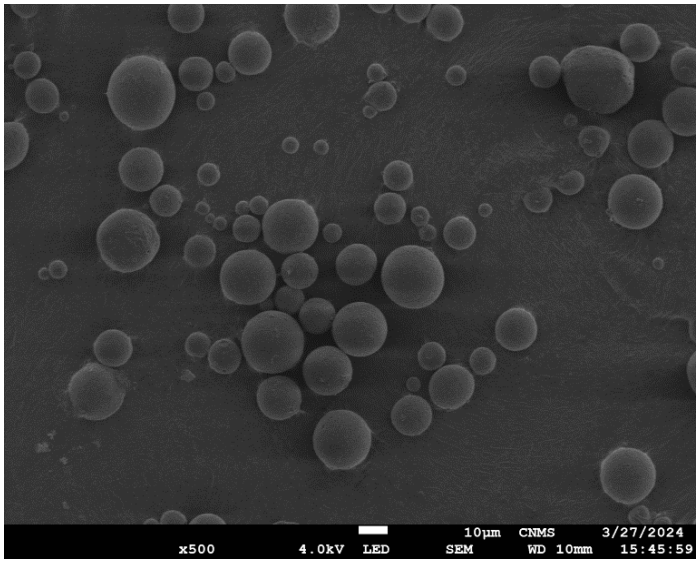
Following the preparation of nanoparticles via the Solvent-Antisolvent method of nanoprecipitation, a detailed analysis of the surface morphology was performed using field emission scanning electron microscopy (FESEM) to determine particle size. Among the various trials, Trial 2 emerged as a particularly promising formulation, displaying particles whose sizes and shapes closely aligned with the desired criteria. However, a comprehensive evaluation of all trials underscored the need for further optimization. Despite the relative success of Trial 2, the particle sizes observed across all trials ranged from 1 micrometer to 5 micrometers. This variability underscores the necessity of achieving uniform nanometer-scale nanoparticles to meet the desired performance and functionality criteria.

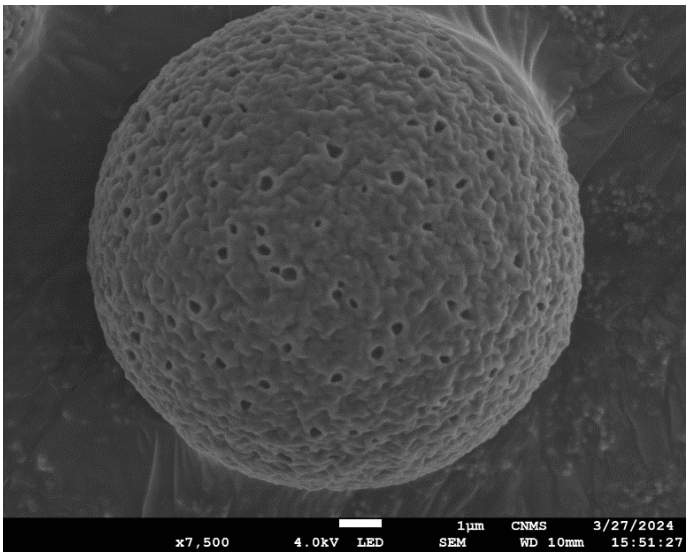
It's important to note that only the first trial involved nanoparticles loaded with Rutin, whereas the subsequent trials (including the promising Trial 2) consisted solely of plain PEG-PLGA nanoparticles without any Rutin loading. This distinction further emphasizes the need for additional refinement to consistently produce nanoparticles within the desired size range for optimal performance and to facilitate future studies. Therefore, while Trial 2 showed potential, significant optimization efforts are required to ensure consistent production of nanoparticles that meet the stringent criteria necessary for advanced applications.

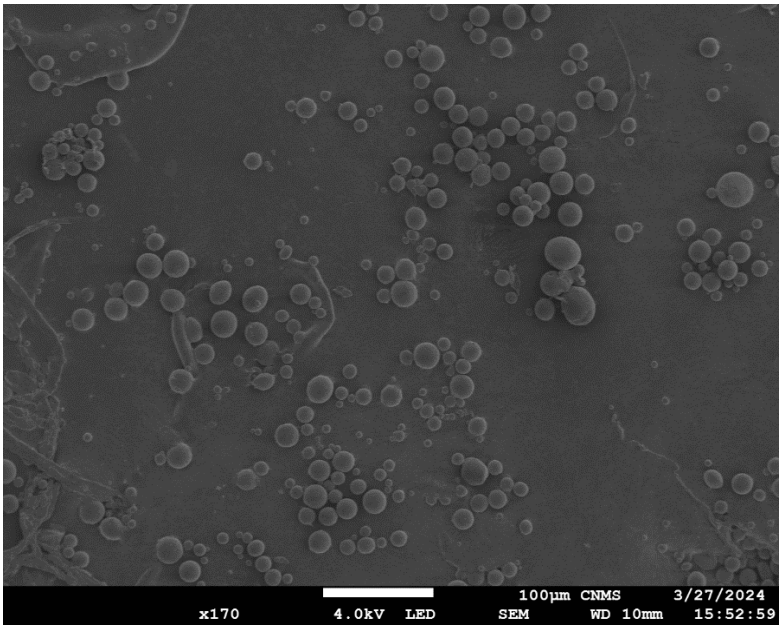

1µm CNMS 3/14/2024
x5,000 2.00kV LED SEM WD 10mm 17:01:19


1µm CNMS 3/14/2024
x20,000 2.00kV LED SEM WD 10mm 17:04:27

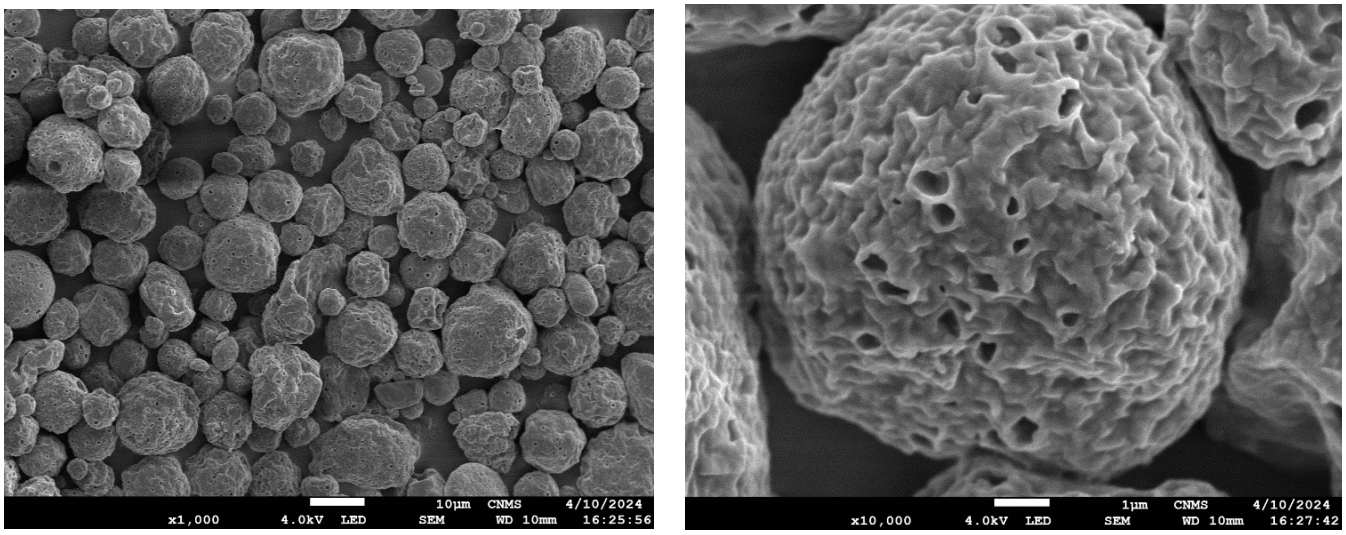
a) TRIAL 1


10µm CNMS 3/27/2024
x500 4.0kV LED SEM WD 10mm 15:45:59

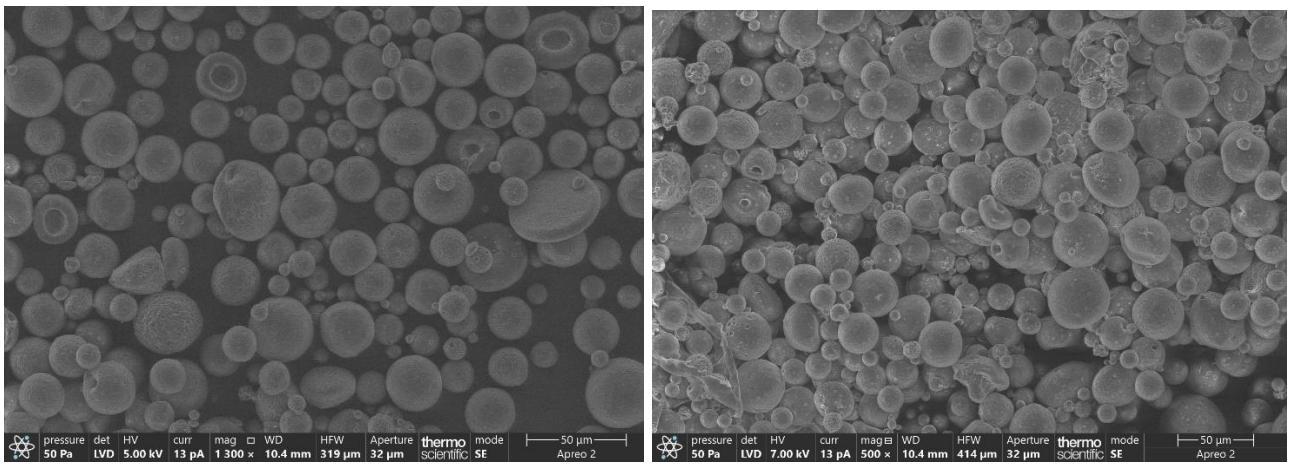

1µm CNMS 3/27/2024
x7,500 4.0kV LED SEM WD 10mm 15:51:27


100µm CNMS 3/27/2024
x170 4.0kV LED SEM WD 10mm 15:52:59

b) TRIAL 2



c) TRIAL 3



d) TRIAL 4

Fig 4.12 FSEM IMAGES OF NANOPARTICLES

CHAPTER 5: SUMMARY AND CONCLUSIONS

This study primarily aimed to comprehensively explore the potential of Rutin-encapsulated PEG-PLGA nanoparticles as a novel therapeutic approach for cancer treatment. The primary objectives were to prepare Rutin-loaded PEG-PLGA nanoparticles, investigate the inhibitory potential of Rutin on proteins involved in autophagy and metabolism pathways through molecular docking. The research mainly focuses on the synthesis of these nanoparticles using the solvent-antisolvent method, followed by their characterization. Important steps in the process include:

1. Nanoparticle Synthesis: Rutin and PEG-PLGA were dissolved in an organic solvent, then introduced into an antisolvent, leading to nanoparticle formation. This method leverages solubility differences to precipitate the nanoparticles.

2. Characterization: The nanoparticles were characterized for their size and shape, with particular attention given to achieving uniform, nanometer-scale particles.

3. Molecular Docking Studies: To understand the potential therapeutic effects of Rutin, molecular docking was performed to assess its inhibitory potential on proteins involved in autophagy and metabolic pathways, comparing these results with known FDA-approved inhibitors.

By integrating nanoparticle development with a deep dive into Rutin's pharmacological activity, the research also aims to provide a comprehensive understanding of its therapeutic potential. The synthesis of nanoparticles using the solvent-antisolvent method allows for precise control over nanoparticle properties by optimizing parameters such as solvent composition, magnetic stirring speed, and solvent to anti-solvent ratio, etc. The resulting nanoparticles are in a way designed to possess desired characteristics, including size, polymeric surface, and drug loading capacity, all crucial for effective drug delivery. Transitioning to the pharmacological evaluation of Rutin, this study also uses molecular docking simulations to explore and explain its interactions with key proteins implicated in cancer progression. Specifically, Rutin's ability to regulate critical cellular pathways, such as autophagy and metabolism, is scrutinized. Autophagy, the process by which cells degrade and recycle their components, and metabolism, the biochemical pathways governing energy production and utilization, are intricately linked to cancer development and progression. Rutin, as a natural flavonoid compound, possesses properties that make it a potential

modulator of these pathways. By assessing Rutin's binding affinity and potential inhibitory effects on proteins involved in these pathways, the study sheds light on its capacity to disrupt cancer cell processes.

Moreover, the study embraces a translational perspective by bridging nanotechnology with molecular docking, thereby laying the groundwork for future clinical applications. The combination of advanced nanoparticle engineering techniques with cutting-edge computational modelling provides a robust platform for the design and evaluation of innovative cancer therapeutics. Beyond the laboratory setting, the potential of Rutin-loaded nanoparticles to overcome existing limitations in cancer treatment, such as drug resistance and off-target toxicity, holds promise for improving patient outcomes. It's important to realize that moving from lab discoveries to actual treatments for patients is tricky. We need more studies, both in labs and with real patients, to make sure that using Rutin-loaded PEG-PLGA nanoparticles for cancer treatment is effective, safe, and won't cause harm. We also need to figure out if these nanoparticles work well in different types of cancers and in people. Another thing to think about is if we can make enough of these nanoparticles consistently and get them approved by regulators for medical use. Overall, this study is a big step in finding new ways to treat cancer, combining small particles with medicine to help patients better.

Conclusion

The study's results highlight the exciting potential of PEG-PLGA nanoparticles as a versatile tool in cancer treatment. These nanoparticles offer a remarkable level of flexibility, allowing researchers to customize their properties such as size and surface to meet specific therapeutic needs. By carefully fine-tuning the parameters of the solvent-antisolvent method, Rutin-loaded nanoparticles can be synthesized with precise control over their size and uniformity, which is crucial for effective drug delivery. Moreover, the investigation into Rutin's role as a potential cancer therapeutic revealed promising finding. Rutin, a natural compound found in plants, demonstrated significant binding affinity with key proteins involved in critical cancer pathways, particularly autophagy and metabolism. Utilizing molecular docking analysis, we gained valuable insights into Rutin's potential effectiveness compared to traditional inhibitors targeting the same proteins. These findings suggest that Rutin could serve as a novel and promising strategy for cancer treatment. However, despite these promising initial results, it's essential to recognize the need for further research to validate

and expand upon these findings. Additional in vitro studies, conducted in controlled laboratory settings on cancer cell lines, as well as in vivo experiments using animal models, will be necessary to thoroughly evaluate the efficacy and safety profile of Rutin-loaded PEG-PLGA nanoparticles in cancer therapy. These comprehensive investigations will provide crucial insights into the translational potential of this innovative approach and its eventual application in clinical settings for treating cancer in the future.

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